

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Name	Buvanesh Y
Register Number	192372367

MODEL PRACTICAL EXAMINATION – SET 4

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application to run a C program to display the student information using any Cloud Service Provider to demonstrate SaaS, with the template of Student Name, Register No., Address, Phone, Age, Courses, Grades and Attendance Report, Progress Report, Semester Mark Sheet forms.
2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.
3. Create a Clone of a VM and test it by loading the previous version/cloned VM.
4. Install a C compiler in the virtual machine created using VirtualBox/VMware Workstation/Player and execute simple C programs.

EXP NO 1: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR STUDENT INFORMATION SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

AIM:

To create a simple cloud software application for student information system using any cloud service provider to demonstrate SaaS.

PROCEDURE:

Step 1: Go to zoho.com.

Step 2: Log into zoho.com.

Step 3: Select Zoho Creator application.

Step 4: Enter application name as Student Information System.

Step 5: Created new application Student Information System.

Step 6: Select create form option.

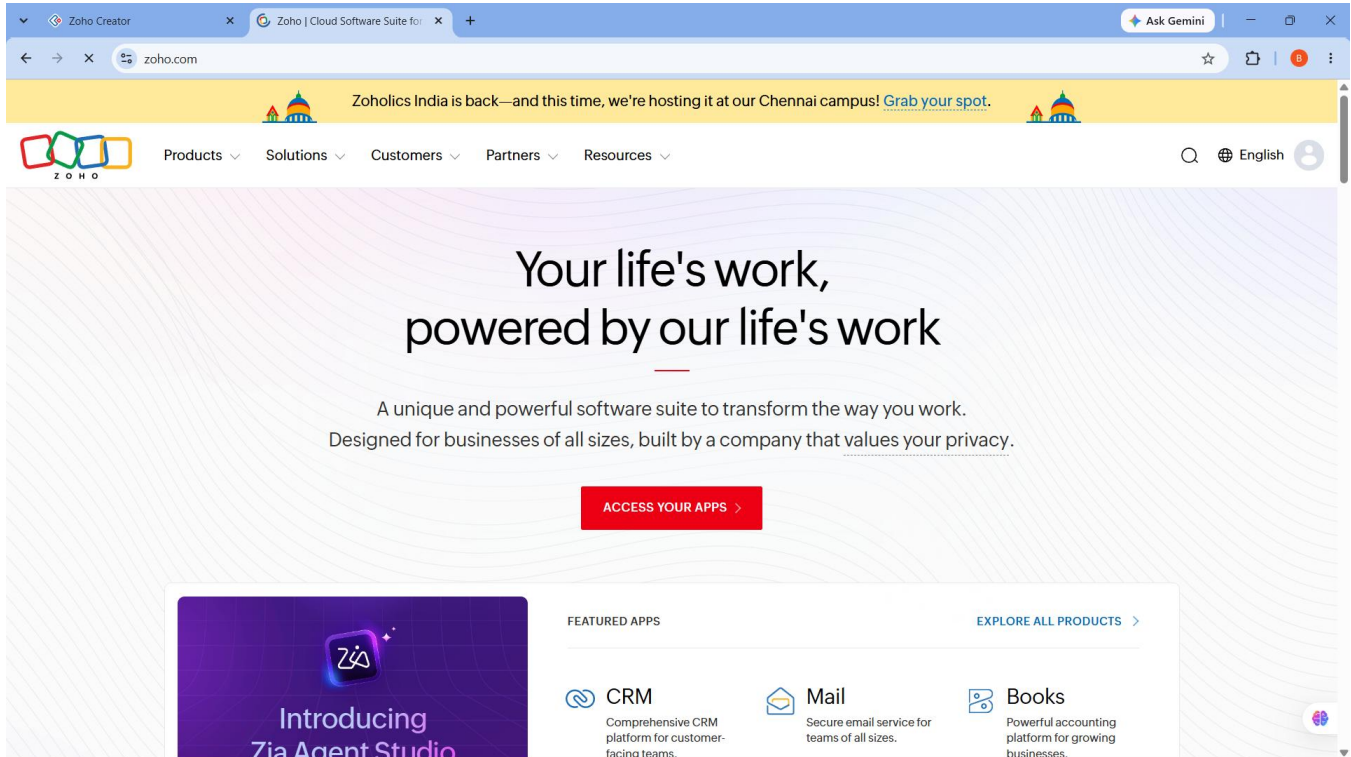
Step 7: Create a form named "Student Details" with fields: Student Name, Register Number, Address, Phone Number, Age, Courses, Grade, Attendance Percentage.

Step 8: Create report pages for Attendance Report, Progress Report, and Semester Mark Sheet using the form data.

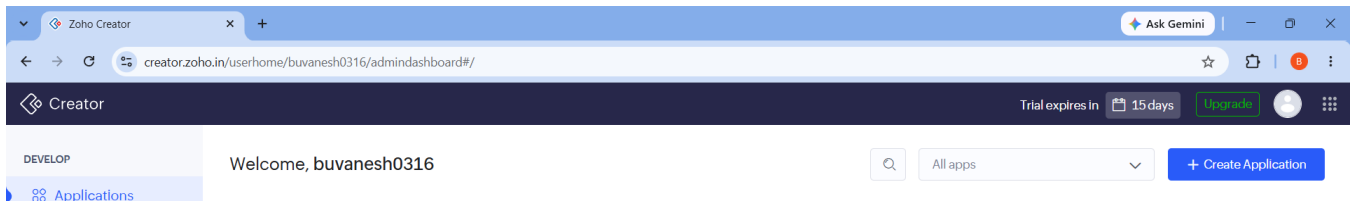
Step 9: The software has been created.

IMPLEMENTATION:

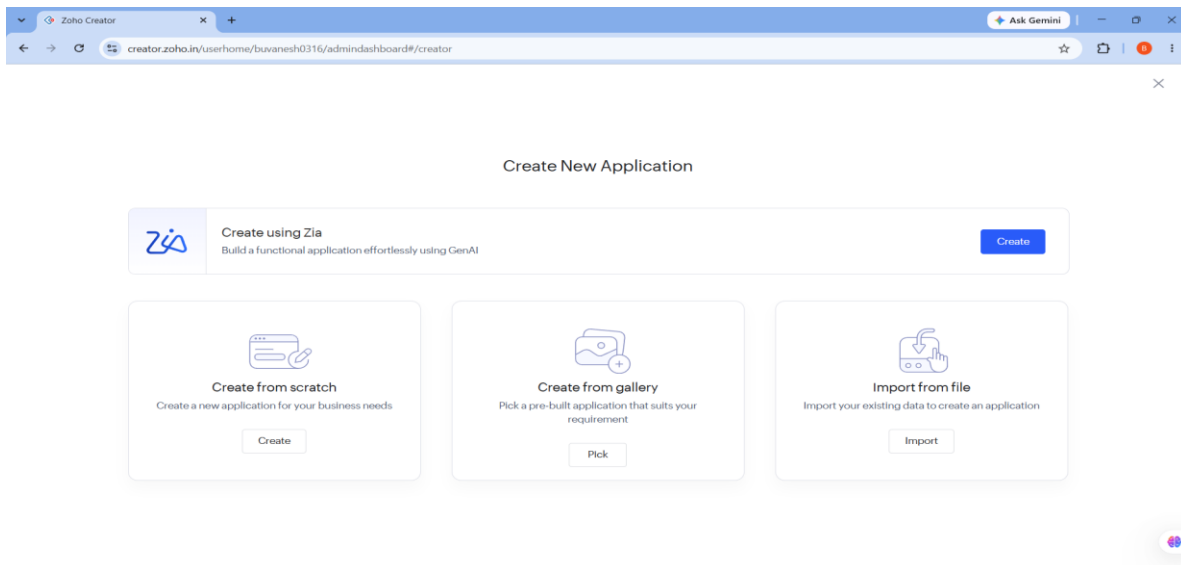
STEP1: GOTO ZOHOCOM



STEP 2: LOGINTOTHE ZOHOCOM

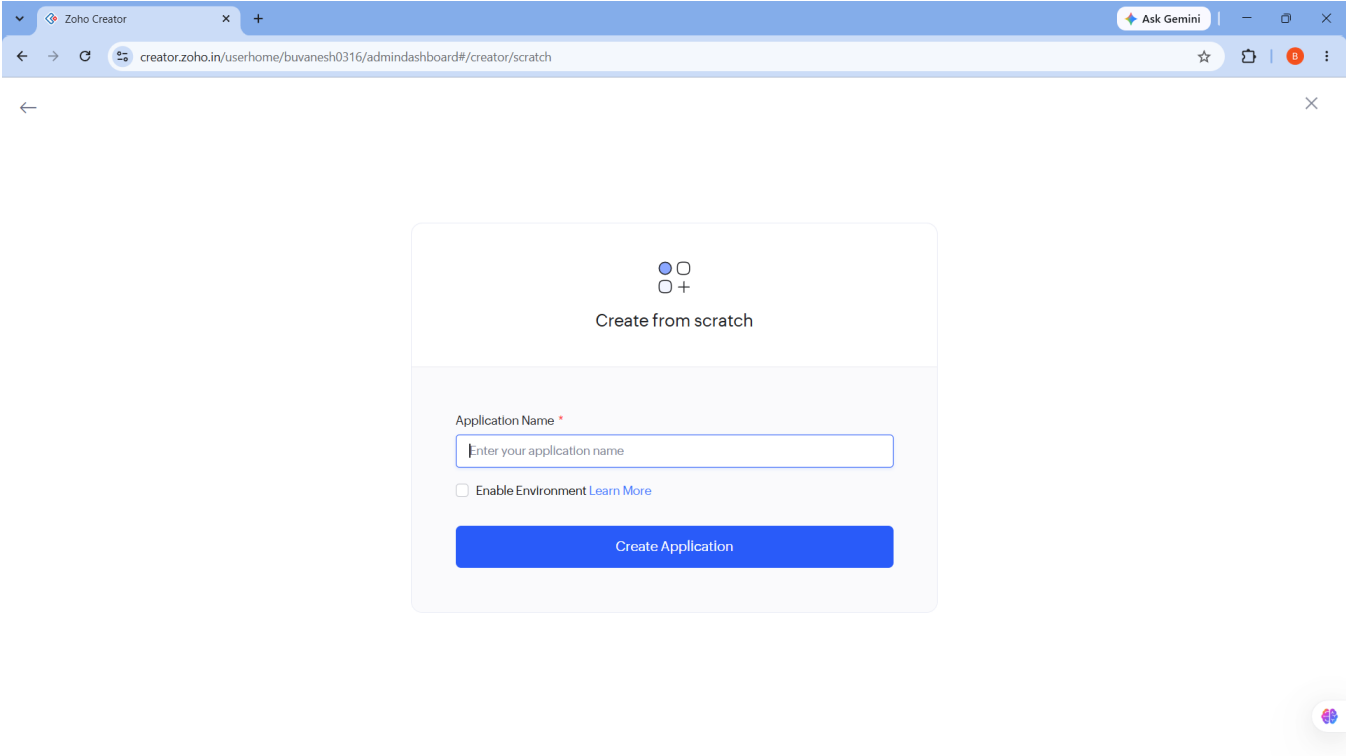


STEP 3: SELECT ONE APPLICATION

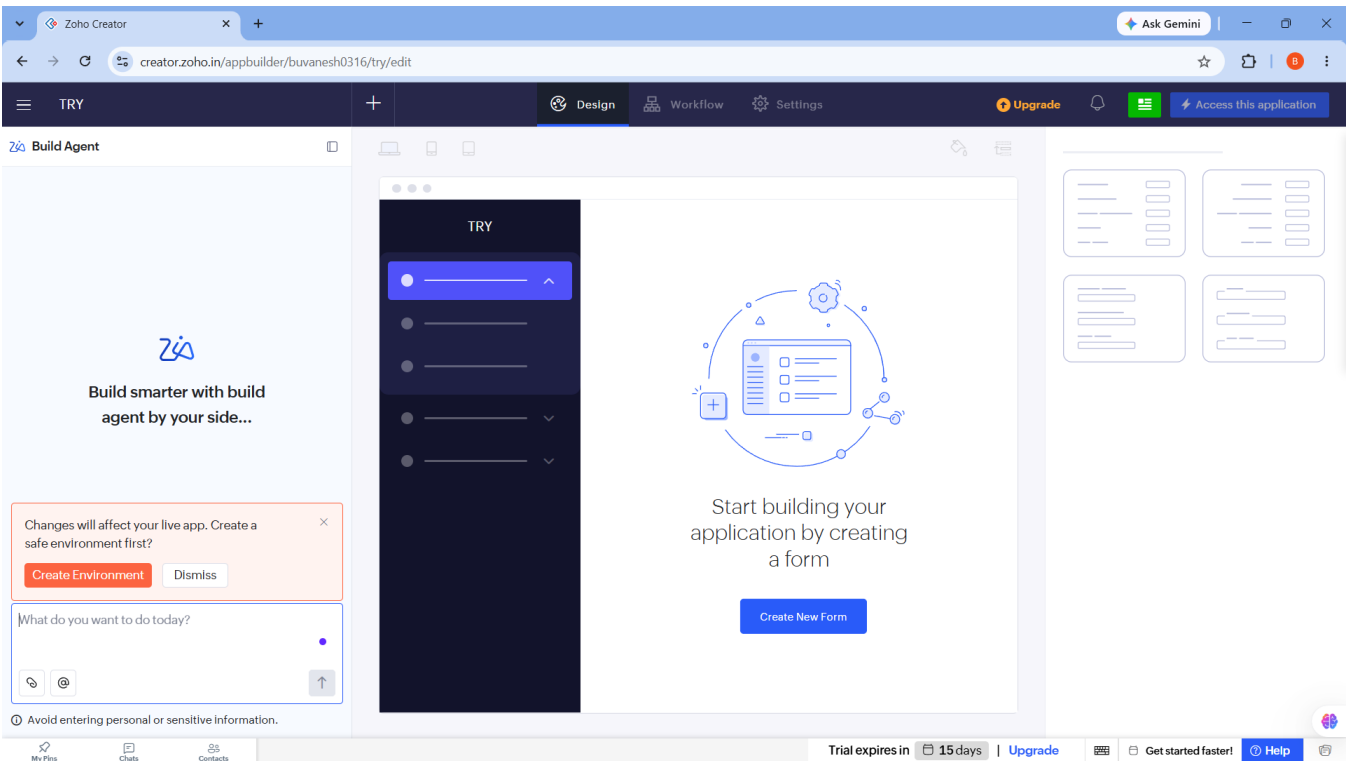


STEP 4: SELECT CREATE FROM SCRATCH

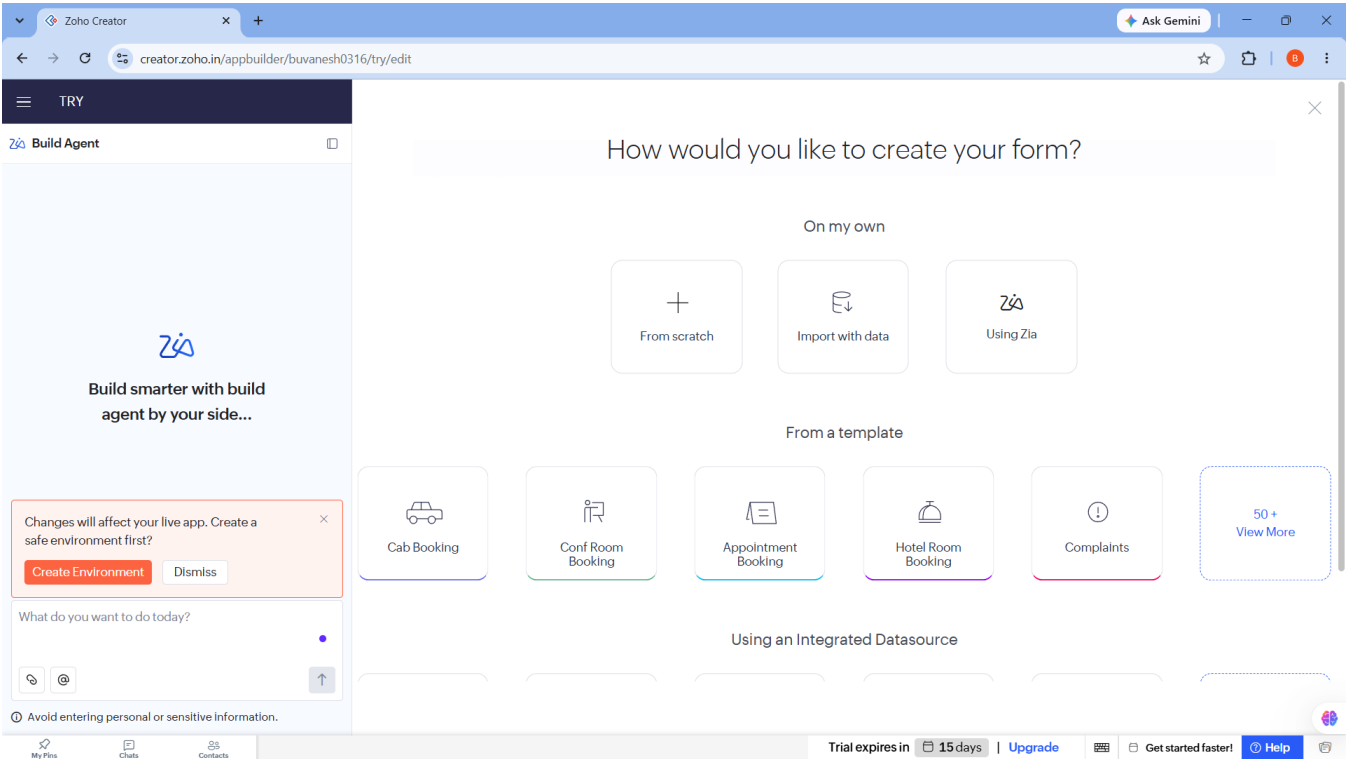
STEP 5: ENTER APPLICATION NAME



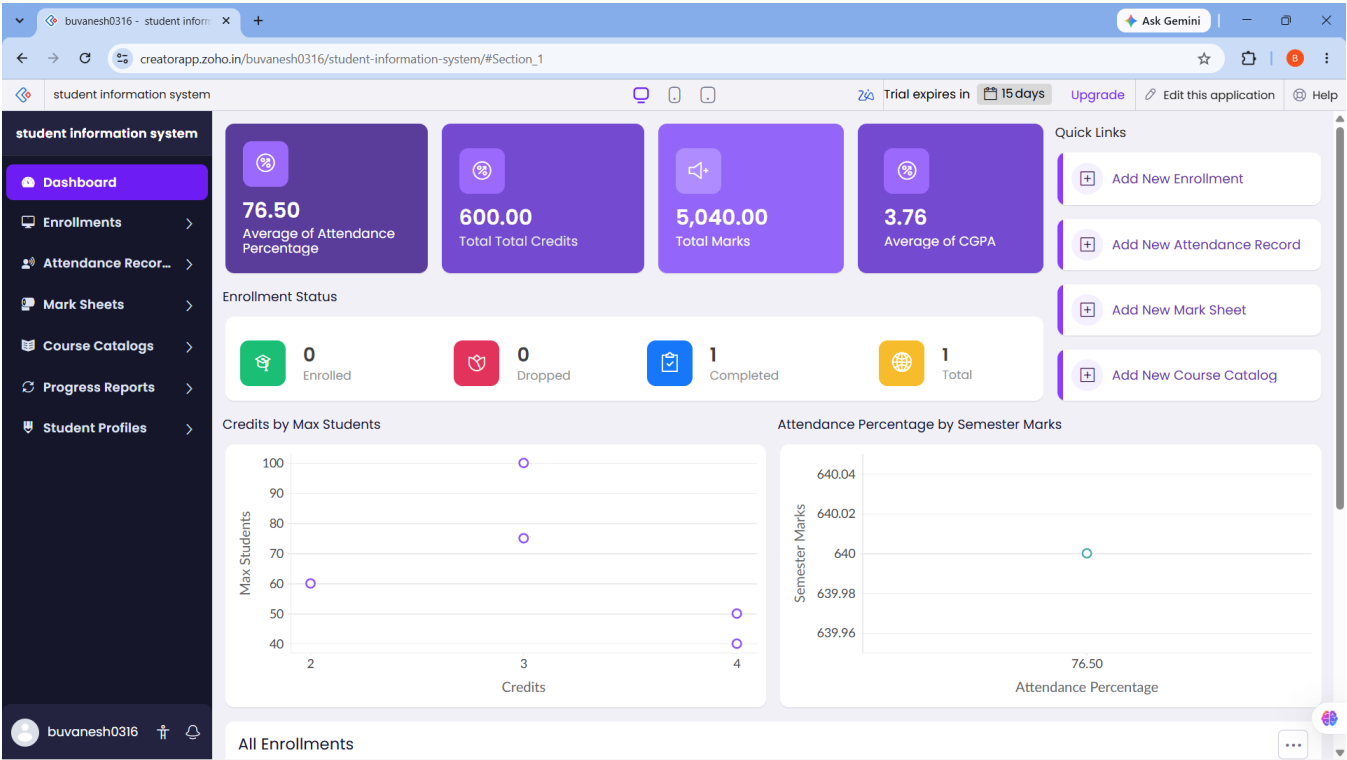
STEP 6: CREATED NEW APPLICATION

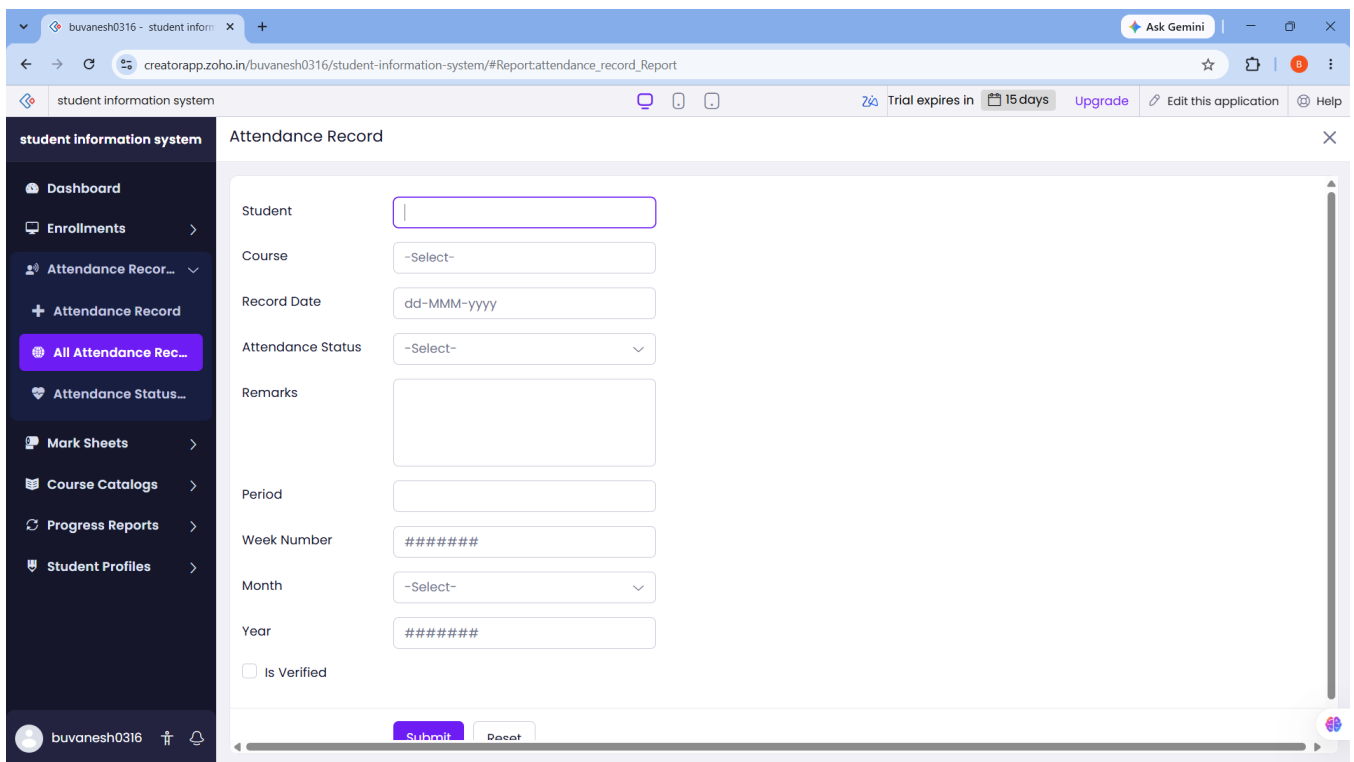
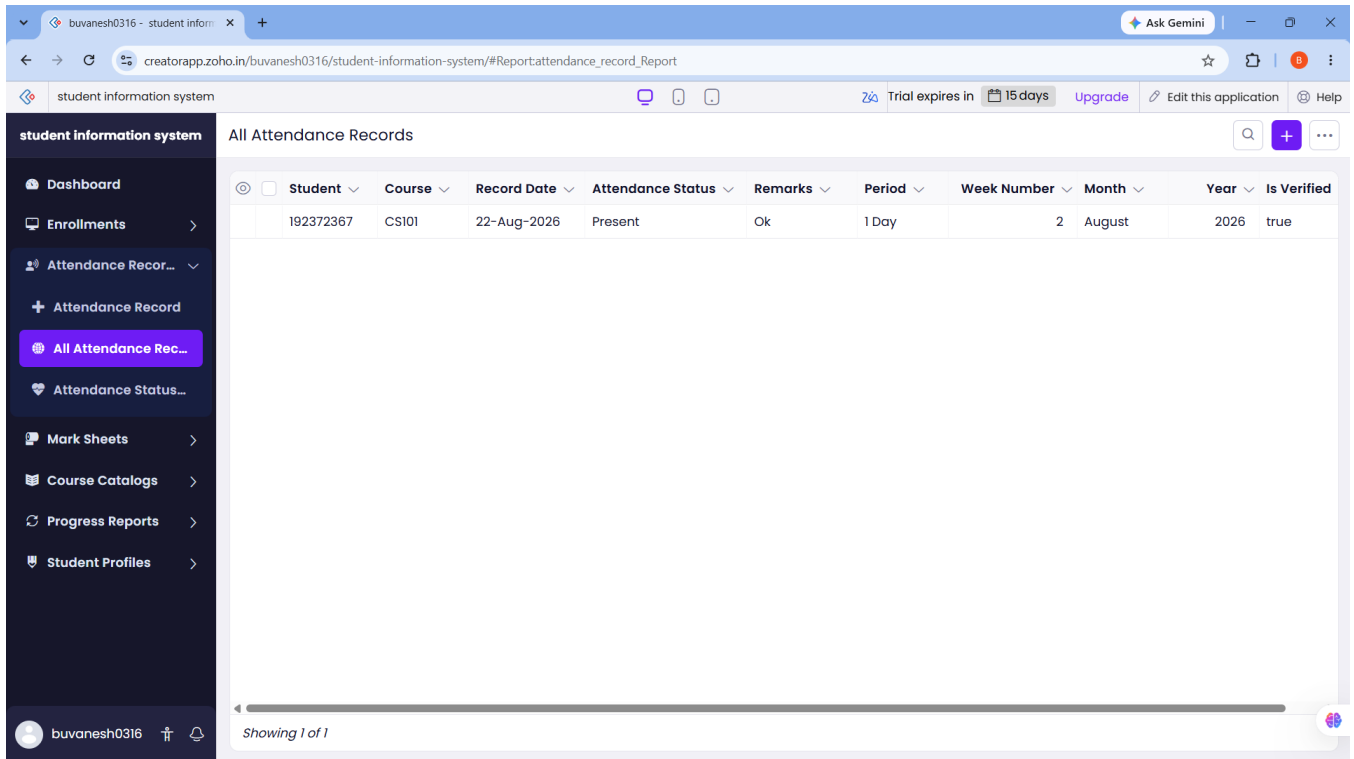


STEP 7: SELECT ONE FORM



STEP 8: THE SOFTWARE HASE BEEN CREATED.





RESULT:

Thus, a simple cloud software application for student information system was successfully created using Zoho Creator to demonstrate SaaS.

EXP NO 2: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR MARK SHEET PROCESSING SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

AIM:

To create a simple cloud software application for mark sheet processing system using any cloud service provider to demonstrate SaaS.

PROCEDURE:

Step 1: Go to zoho.com.

Step 2: Log into zoho.com.

Step 3: Select Zoho Creator application.

Step 4: Enter application name as Mark Sheet Processing System.

Step 5: Created new application Mark Sheet Processing System.

Step 6: Select create form option.

Step 7: Create a form named "Mark Details" with fields: Name of the Student, Register Number, Subjects, Marks, Average, Percentage, Rank, Overall Performance.

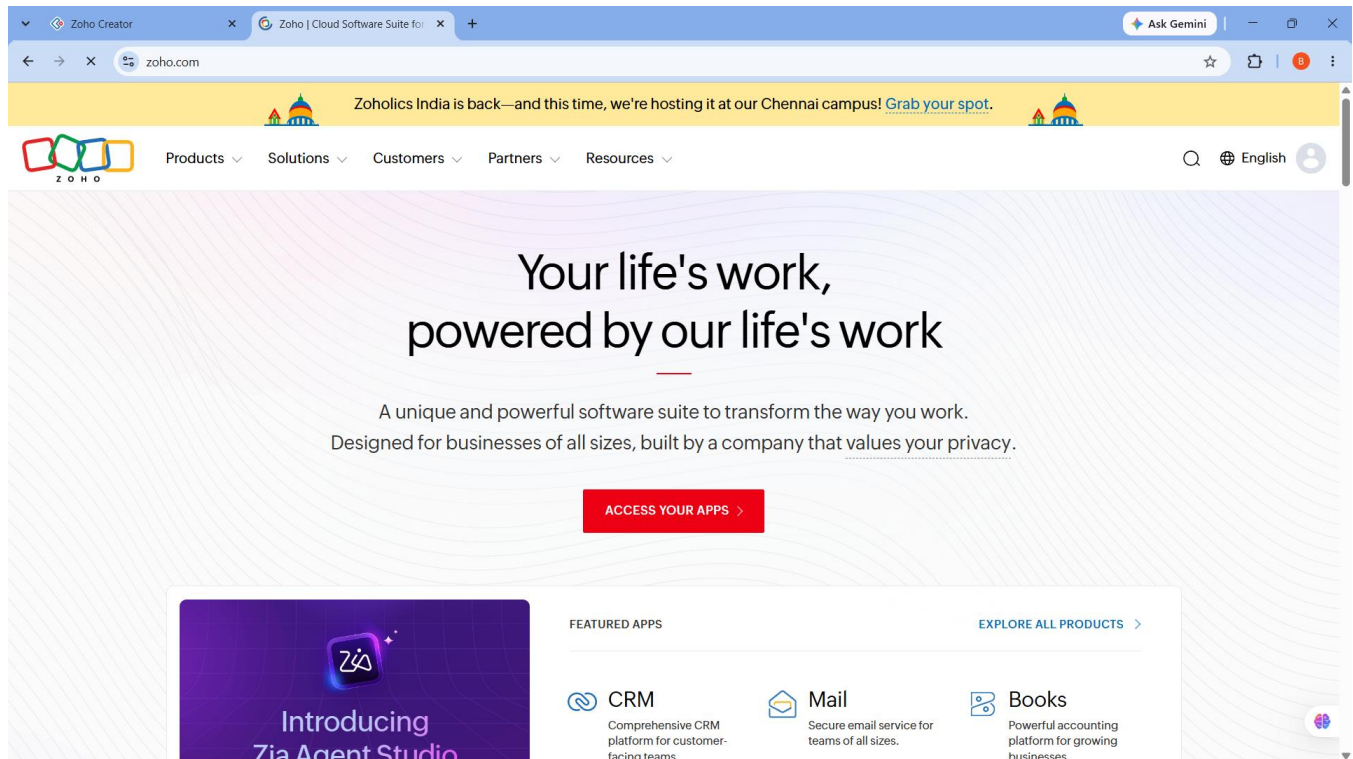
Step 8: Create report pages/deluge functions to auto-calculate Average, Percentage, Rank, and display Overall Performance.

Step 9: The software has been created.

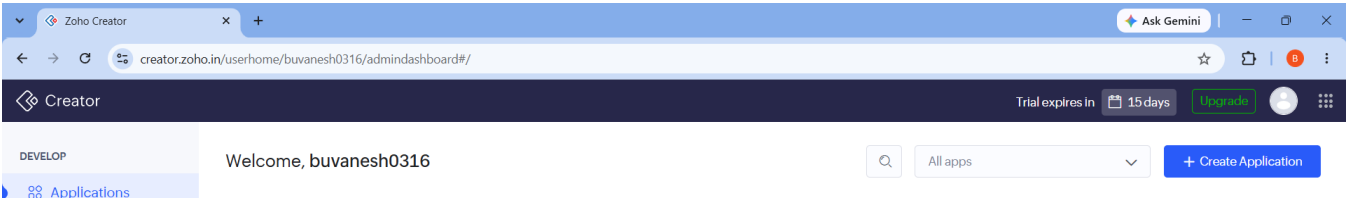
IMPLEMENTATION:

IMPLEMENTATION:

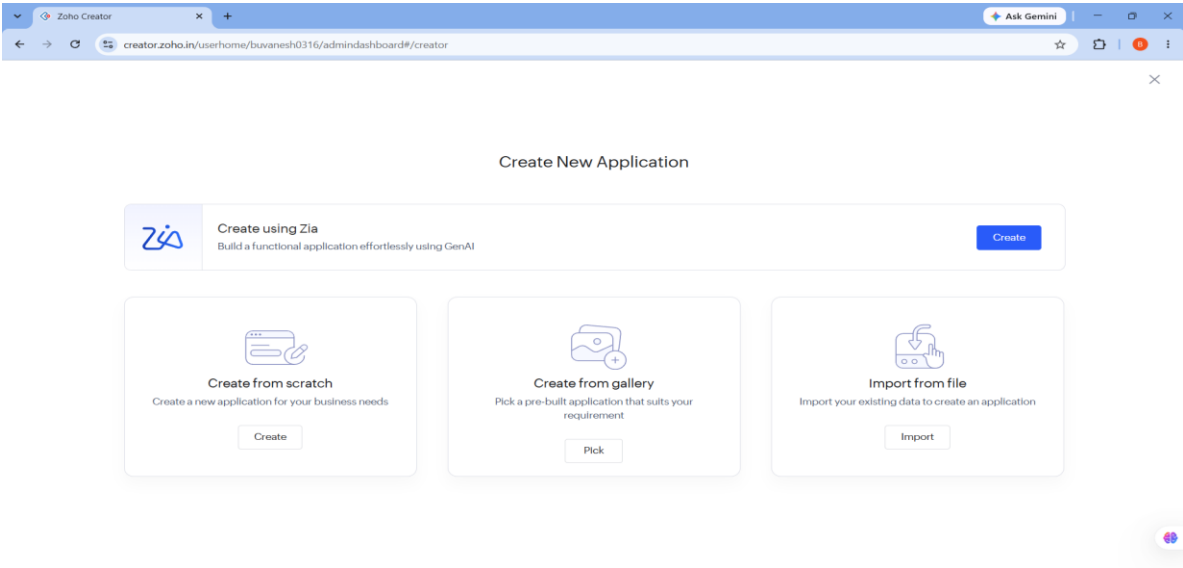
STEP1: GOTO ZOHO.COM



STEP 2: LOGINTOTHE ZOHOCOM

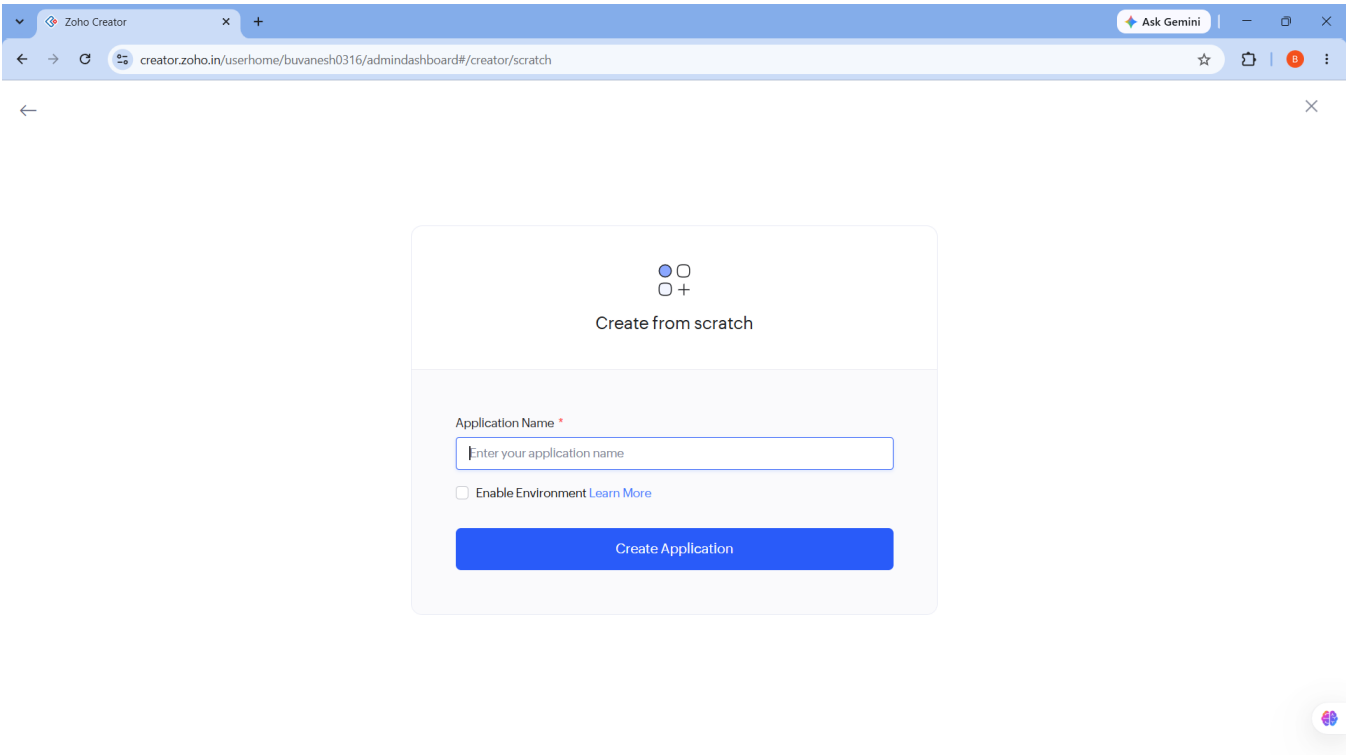


STEP 3: SELECT ONE APPLICATION

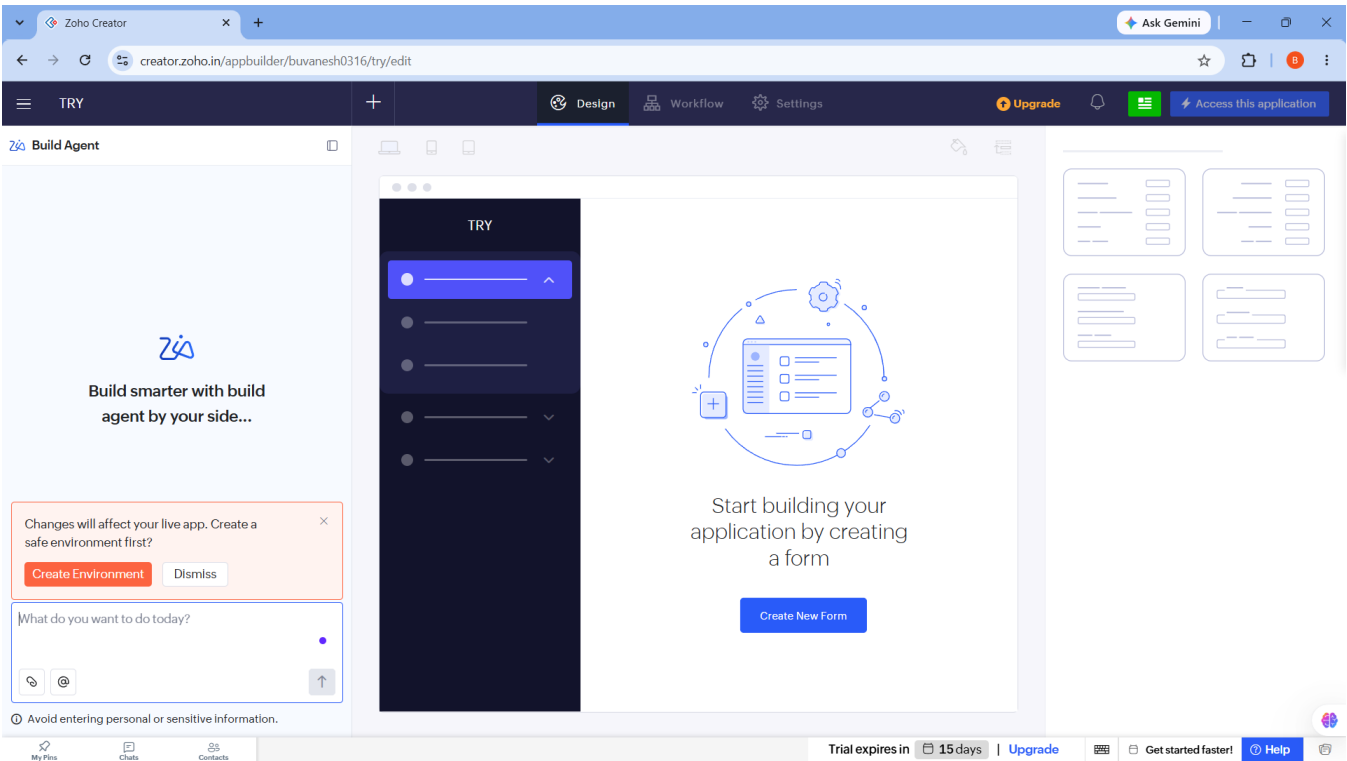


STEP 4:SELECT CREATE FROM SCRATCH

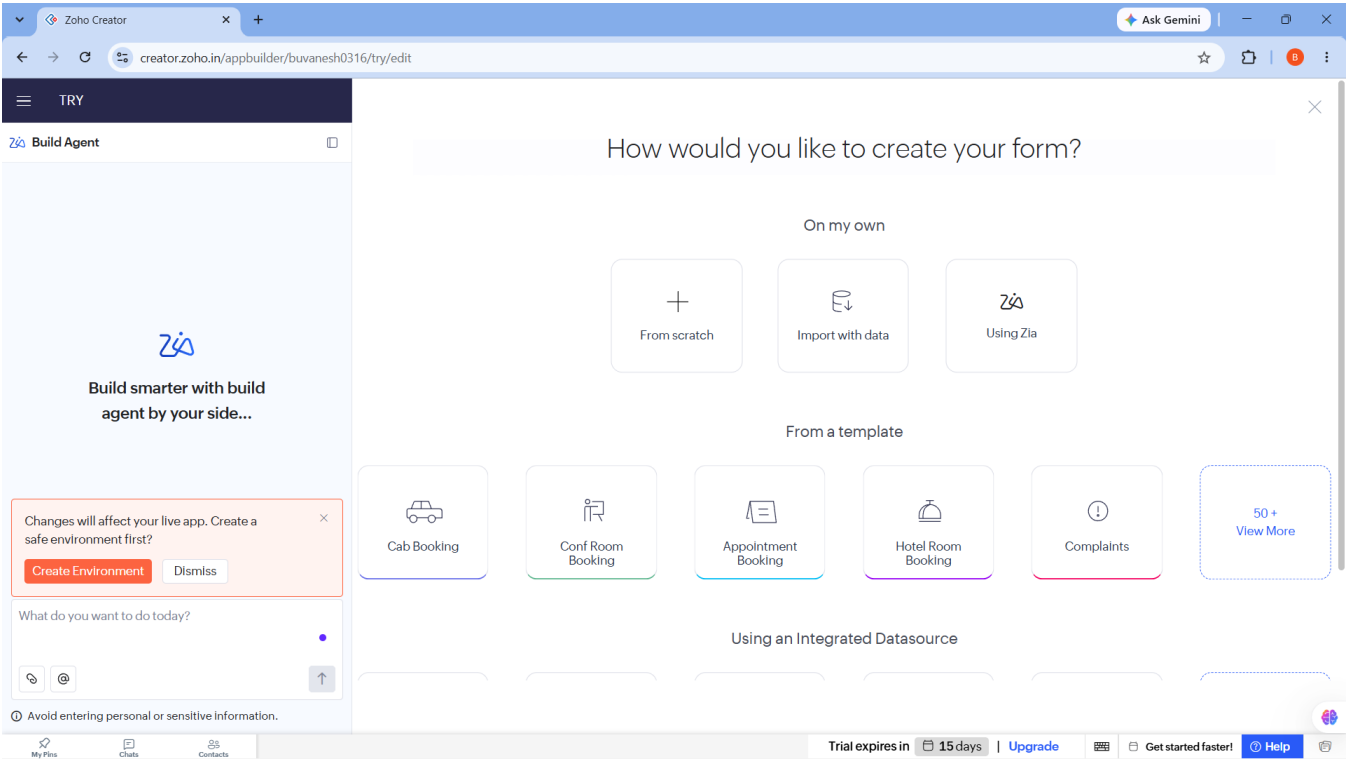
STEP 5: ENTER APPLICATION NAME



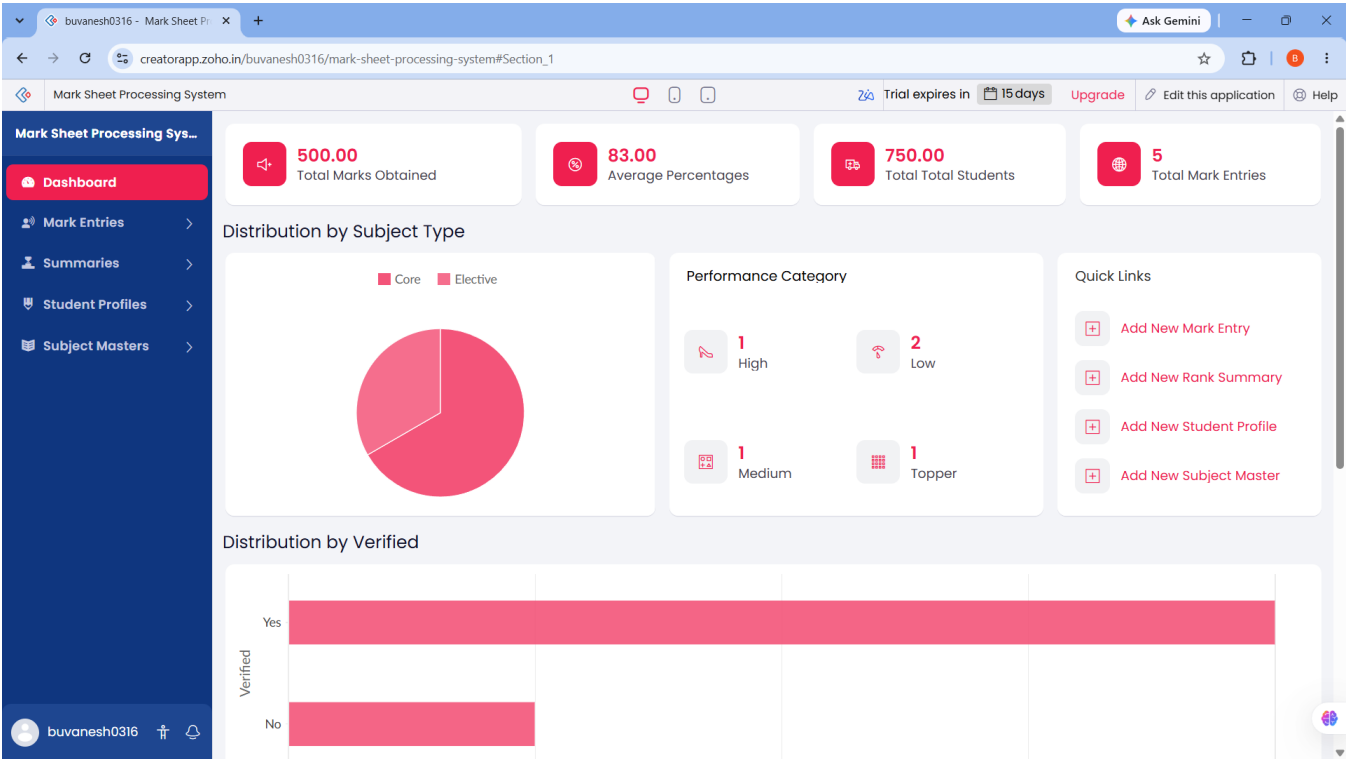
STEP 6: CREATED NEW APPLICATION



STEP 7: SELECT ONE FORM



STEP 8: THE SOFTWARE HASE BEEN CREATED.



Mark Sheet Processing System

All Summaries

Summary Ranked Successfully

Student	Academic Year	Rank	Total Students	Average Marks	Highest Marks	Lowest Marks	Performance Cate...	Calcu
192372367	2026	1	100	100	100	10	Topper	22-Au

Showing 1 of 1

buvanesh0316 - Mark Sheet P | + | Ask Gemini | - |

creatorapp.zoho.in/buvanesh0316/mark-sheet-processing-system#Report:subject_master_Report

Mark Sheet Processing System | Trial expires in 15 days | Upgrade | Edit this application | Help

Mark Sheet Processing Sys... Subject Master

Dashboard

Mark Entries >

Summaries >

Student Profiles >

Subject Masters >

+ Subject Master

All Subject Masters

Subject Types

buvanesh0316

Subject Code: CSAI522

Subject Name: Cloud Computing

Subject Type: Core

Passing Marks: 2027

Credits: 4

Department: CSE

Semester: 5

☒ Is Active

Description: ONLY For Core CSE Students

Submit Reset

RESULT:

Thus, a simple cloud software application for mark sheet processing system was successfully created using Zoho Creator to demonstrate SaaS.

EXP NO 3: CREATE A CLONE OF A VM AND TEST IT BY LOADING THE PREVIOUS VERSION/CLONED VM

AIM:

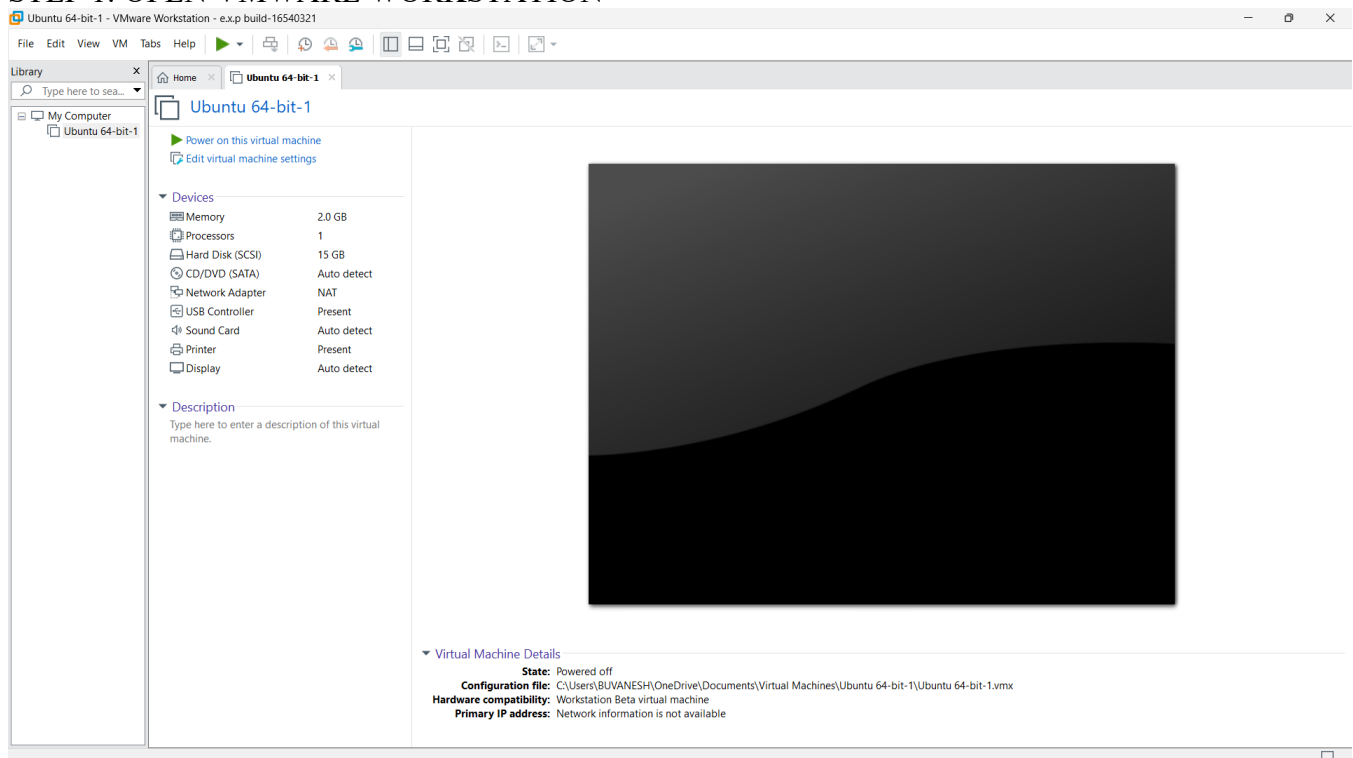
To create a clone of a Virtual Machine and test it by loading the previous version/cloned VM using VMware Workstation.

PROCEDURE:

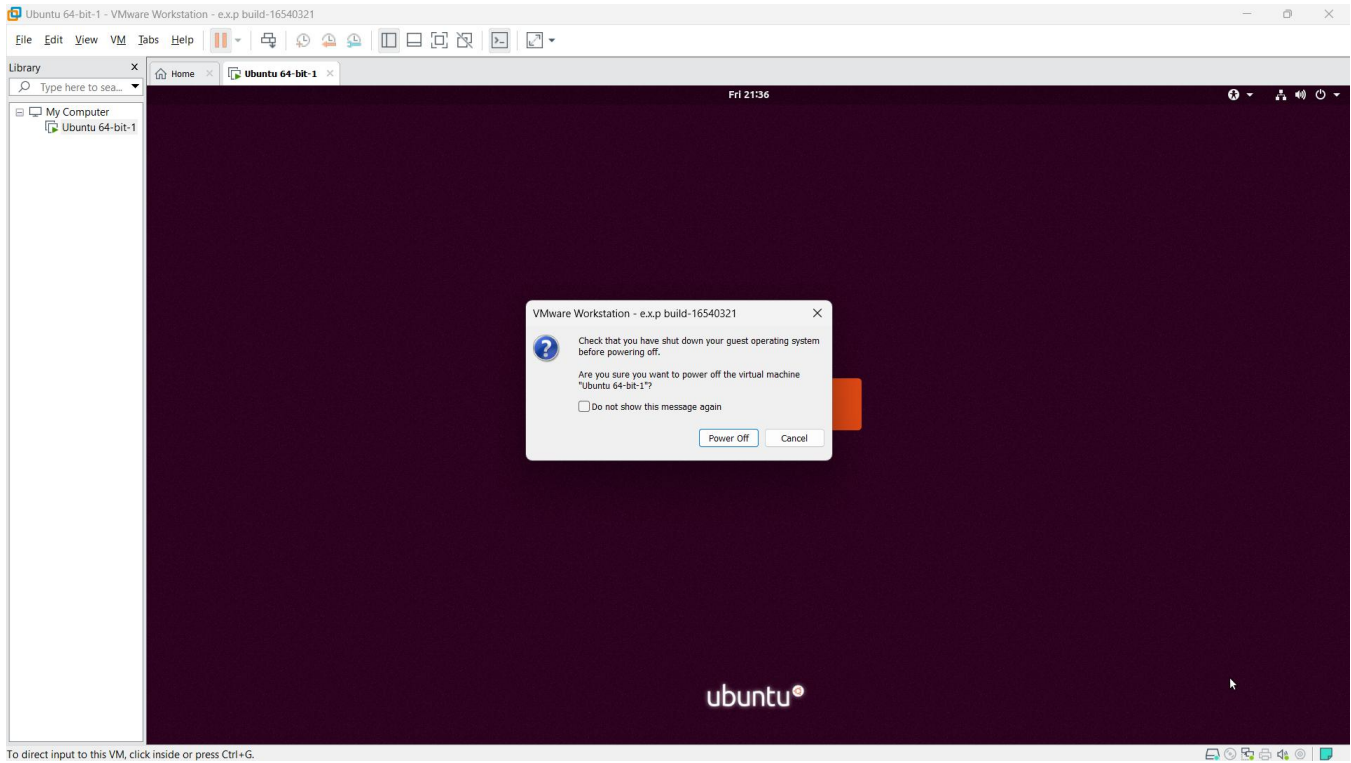
- Step 1: Open VMware Workstation and select the source VM.
- Step 2: Power off the VM completely.
- Step 3: Right-click the VM and select Manage > Clone.
- Step 4: Choose "The current state in the virtual machine" as the clone source.
- Step 5: Select Clone Type as "Create a Full Clone."
- Step 6: Enter a name for the cloned VM and choose the storage location.
- Step 7: Click Finish to complete the cloning process.
- Step 8: Power on the cloned VM and verify it boots successfully.
- Step 9: Take a snapshot of the original VM before making changes, to preserve the previous version.
- Step 10: Revert the original VM to the snapshot to test loading the previous version.
- Step 11: Compare the cloned VM and the reverted previous version to confirm both work correctly.

IMPLEMENTATION:

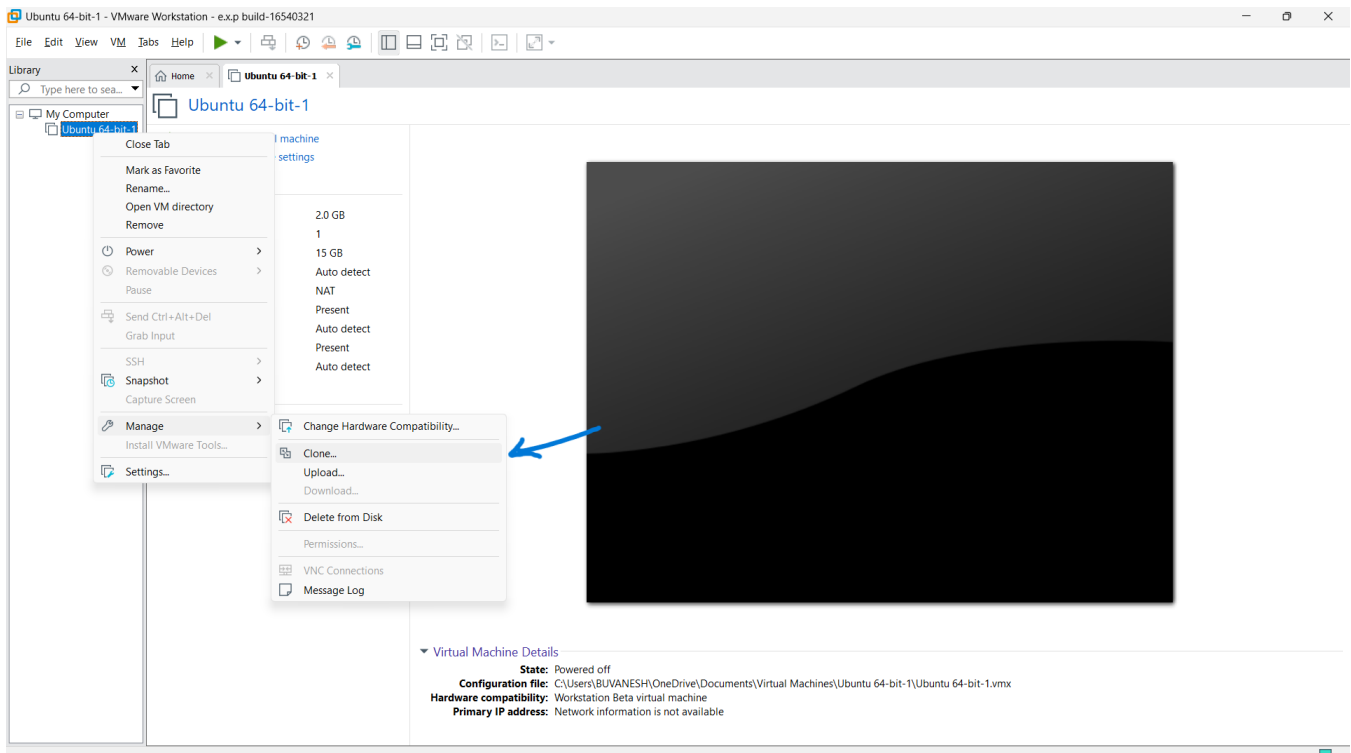
STEP 1: OPEN VMWARE WORKSTATION



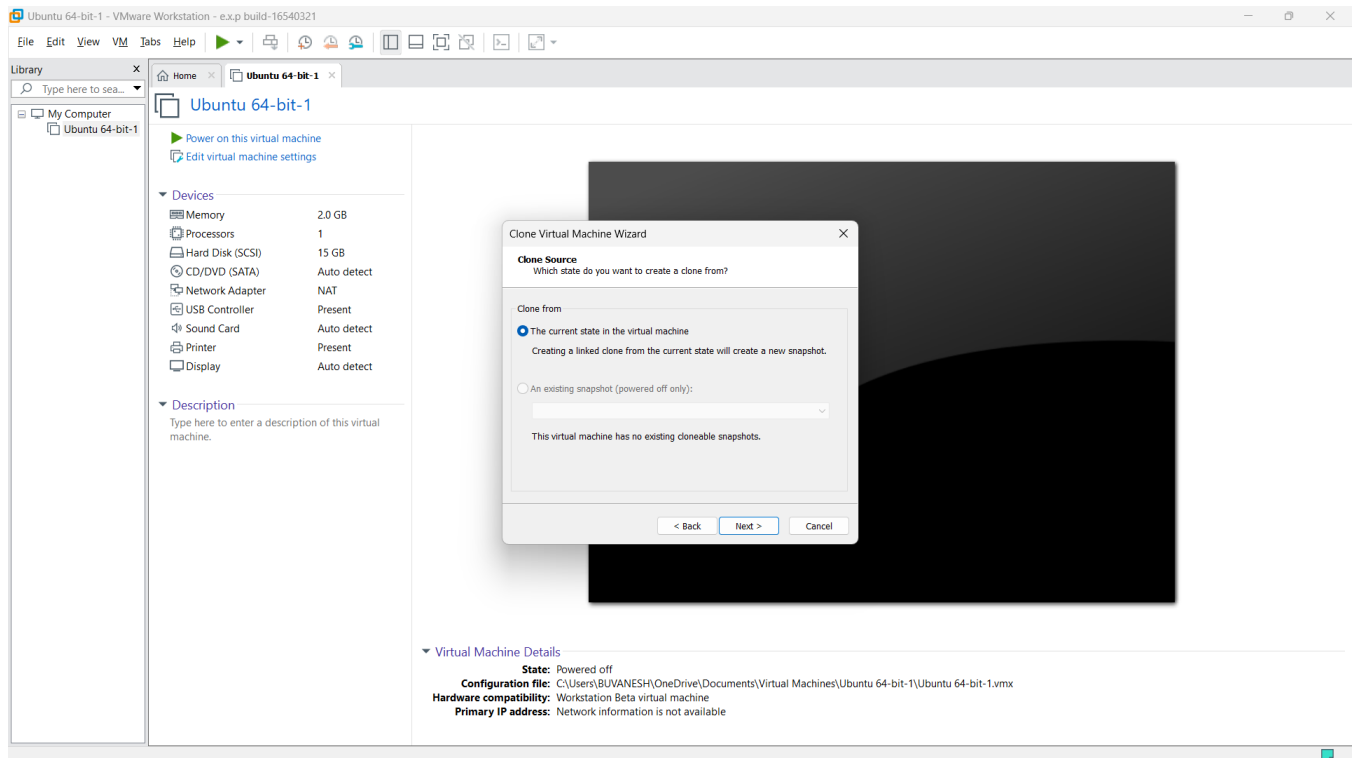
STEP 2: SELECT THE SOURCE VM AND POWER IT OFF



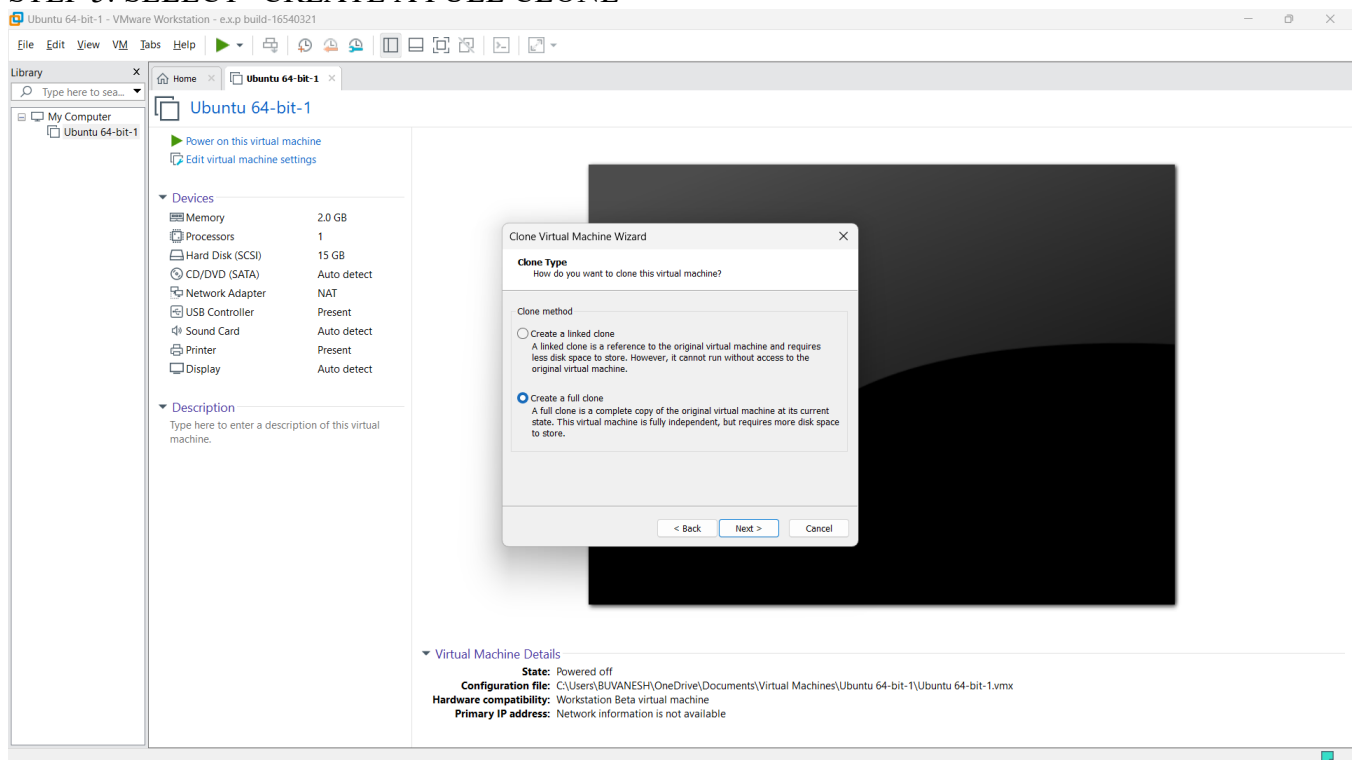
STEP 3: RIGHT-CLICK → MANAGE → CLONE:



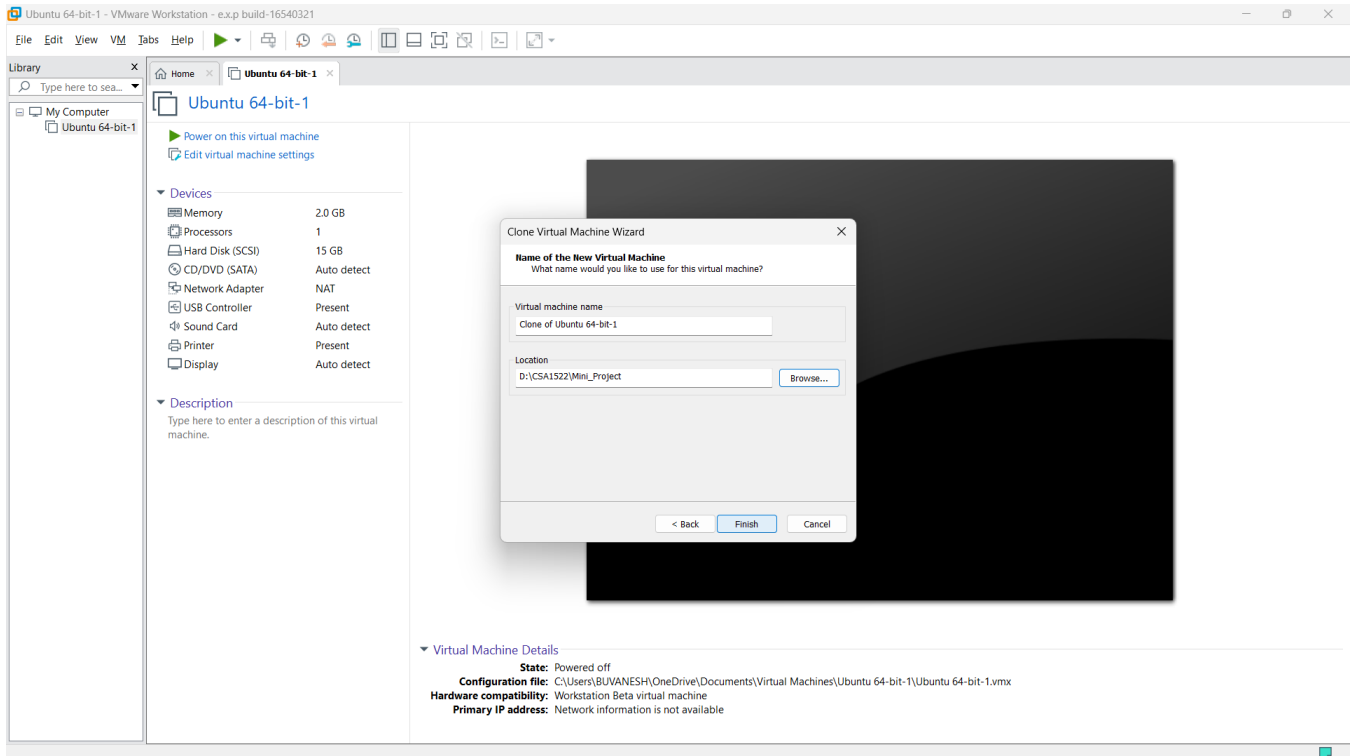
STEP 4: SELECT "CURRENT STATE" AS CLONE SOURCE



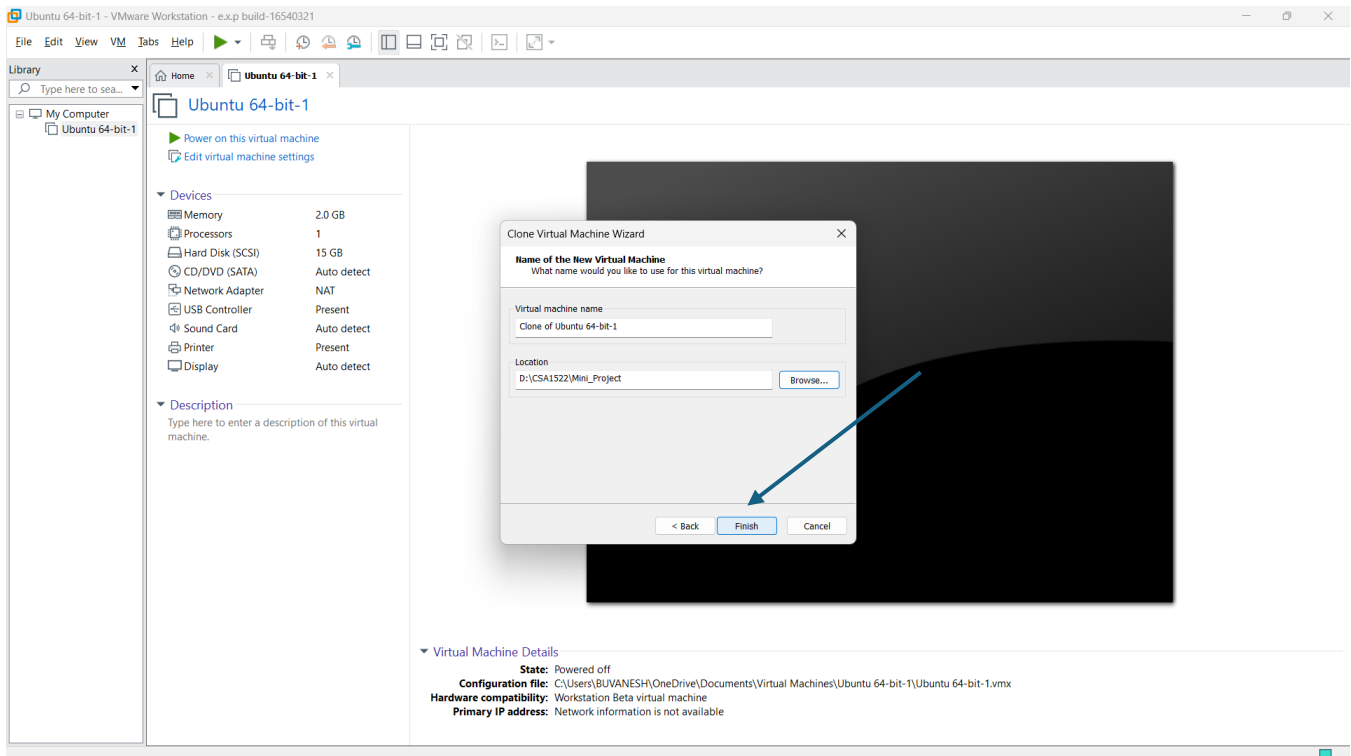
STEP 5: SELECT "CREATE A FULL CLONE"



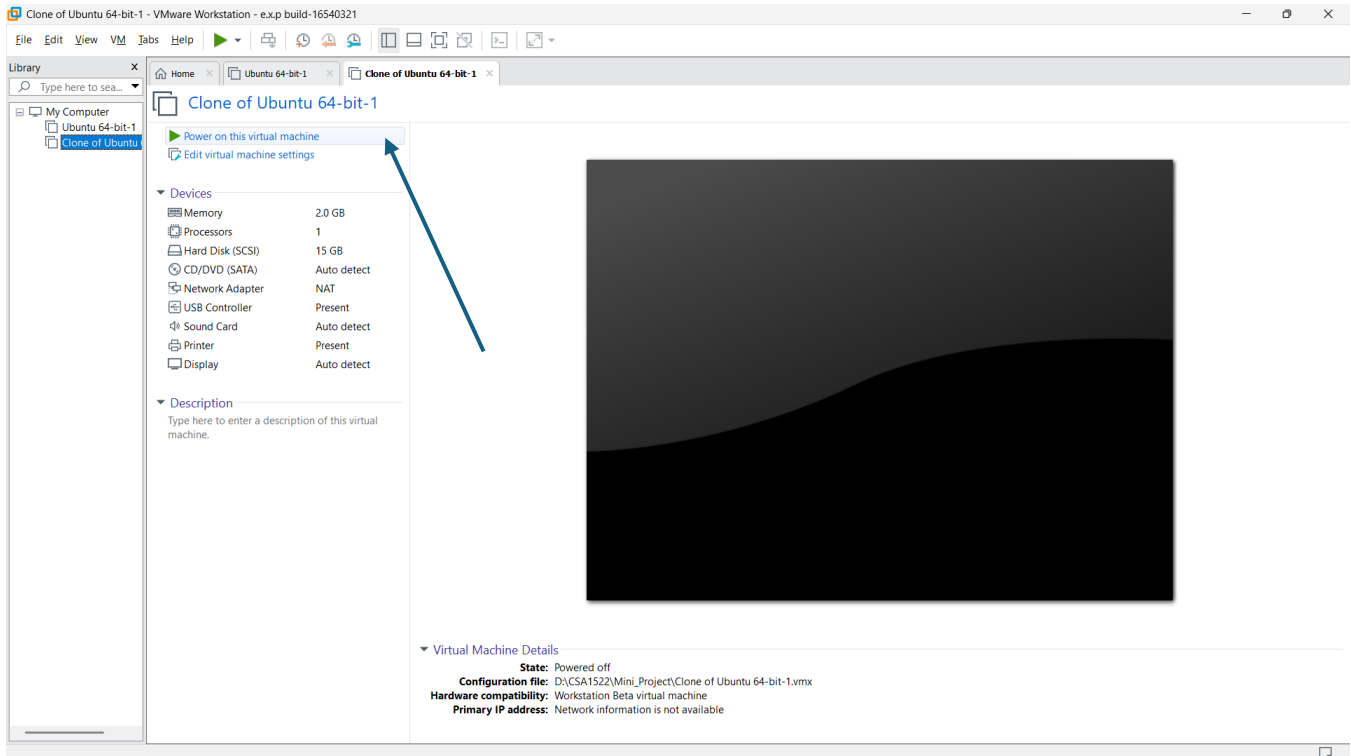
STEP 6: ENTER CLONE NAME AND STORAGE PATH



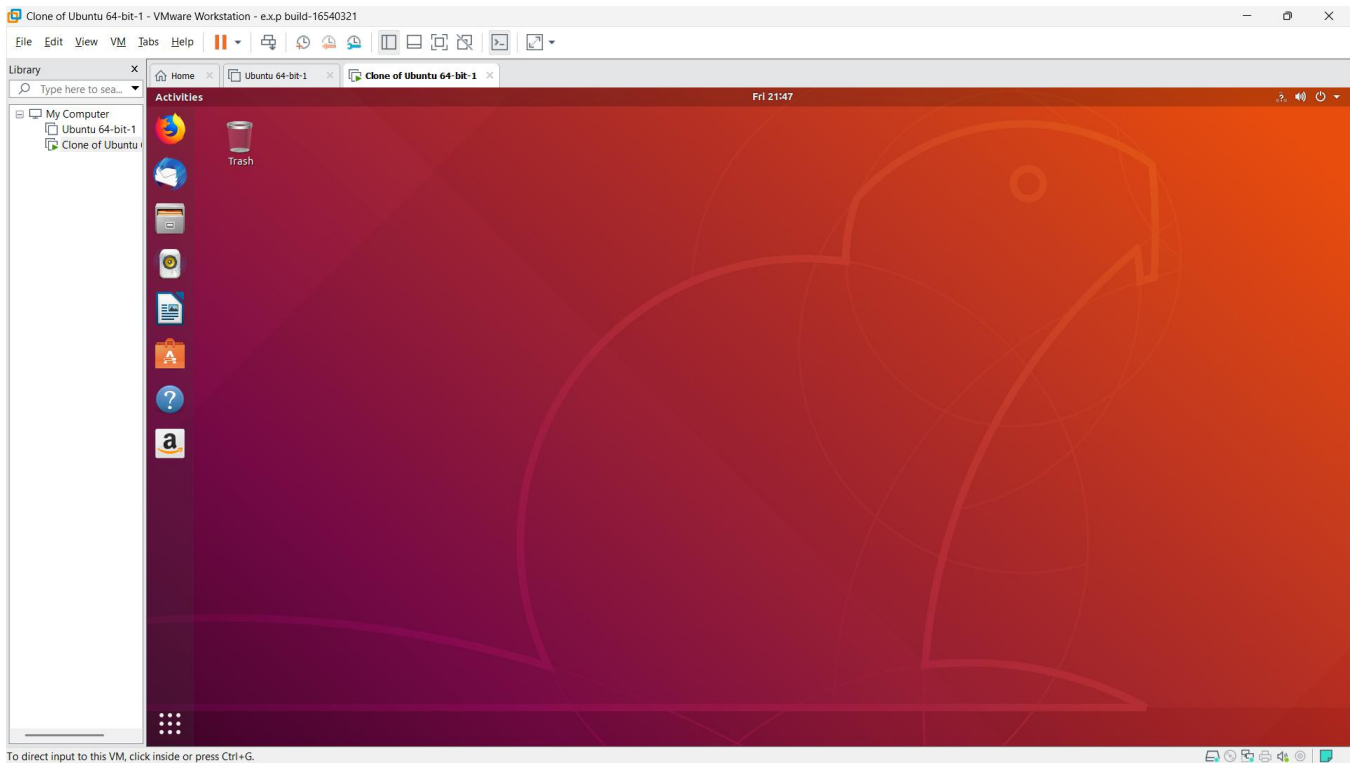
STEP 7: CLICK FINISH TO CREATE THE CLONE



STEP 8: POWER ON THE CLONED VM AND VERIFY



STEP 9: CLONED VM



RESULT:

Thus, a clone of a VM was successfully created and tested by loading the previous version/cloned VM using VMware Workstation.

EXP NO 4: INSTALL A C COMPILER IN THE VIRTUAL MACHINE CREATED USING VIRTUALBOX/VMWARE WORKSTATION/PLAYER AND EXECUTE SIMPLE C PROGRAMS

AIM:

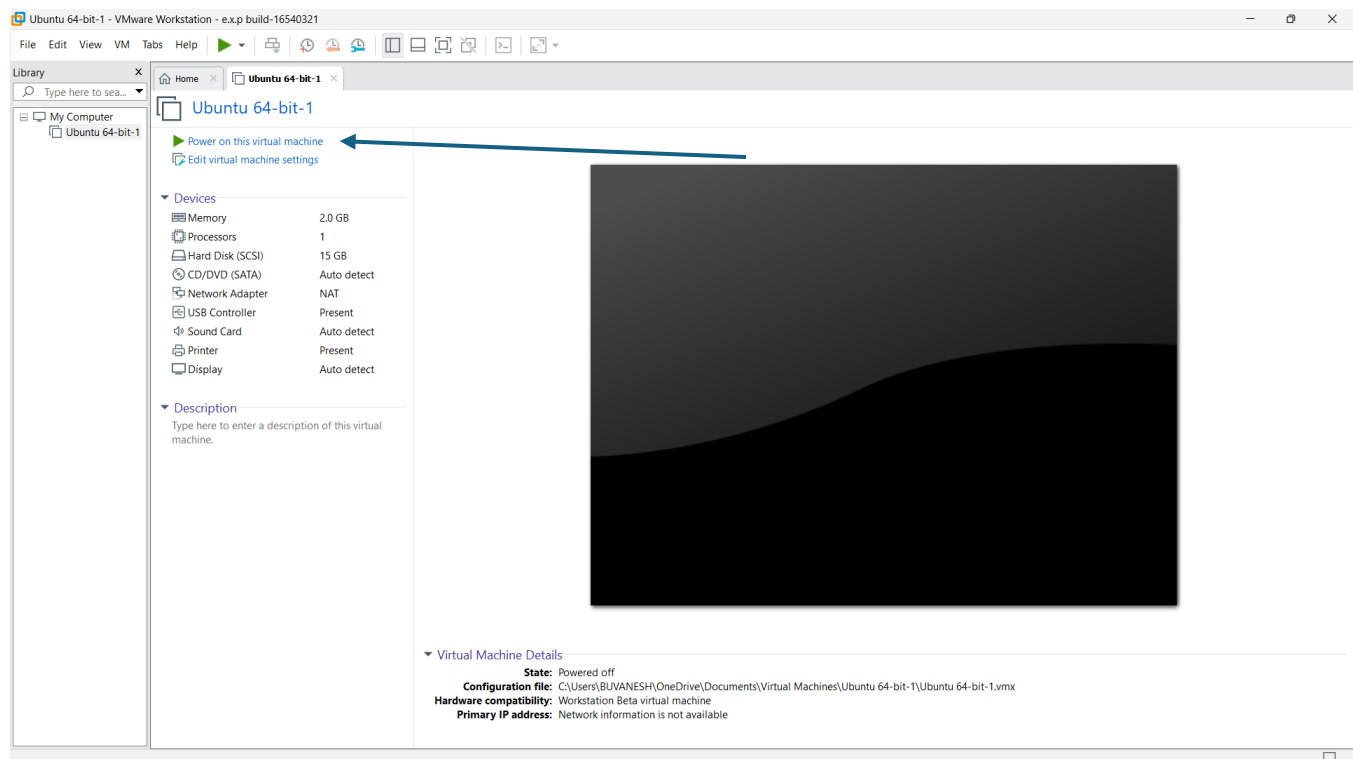
To install a C compiler in the virtual machine created using VirtualBox/VMware Workstation/Player and execute simple C programs.

PROCEDURE:

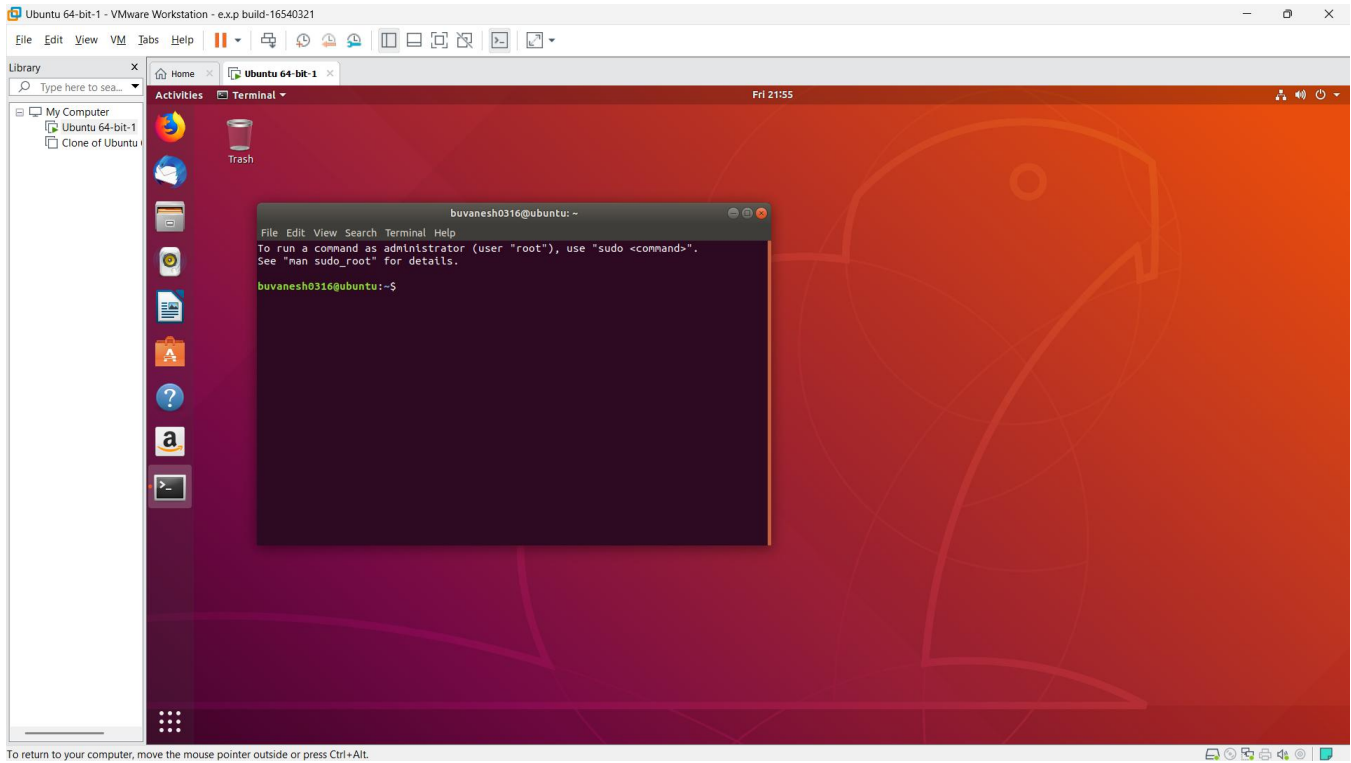
- Step 1: Open VirtualBox/VMware Workstation and power on the virtual machine (e.g., Ubuntu/Linux OS).
- Step 2: Log into the virtual machine with the required credentials.
- Step 3: Open the terminal window inside the VM.
- Step 4: Update the package list using the command: `sudo apt update`.
- Step 5: Install the GCC compiler using the command: `sudo apt install gcc`.
- Step 6: Verify the installation using the command: `gcc --version`.
- Step 7: Open a text editor and write a simple C program (e.g., Hello World).
- Step 8: Save the file with a .c extension (e.g., program.c).
- Step 9: Compile the program using the command: `gcc program.c -o program`.
- Step 10: Execute the compiled program using the command: `./program`.
- Step 11: Verify the output displayed on the terminal.

IMPLEMENTATION:

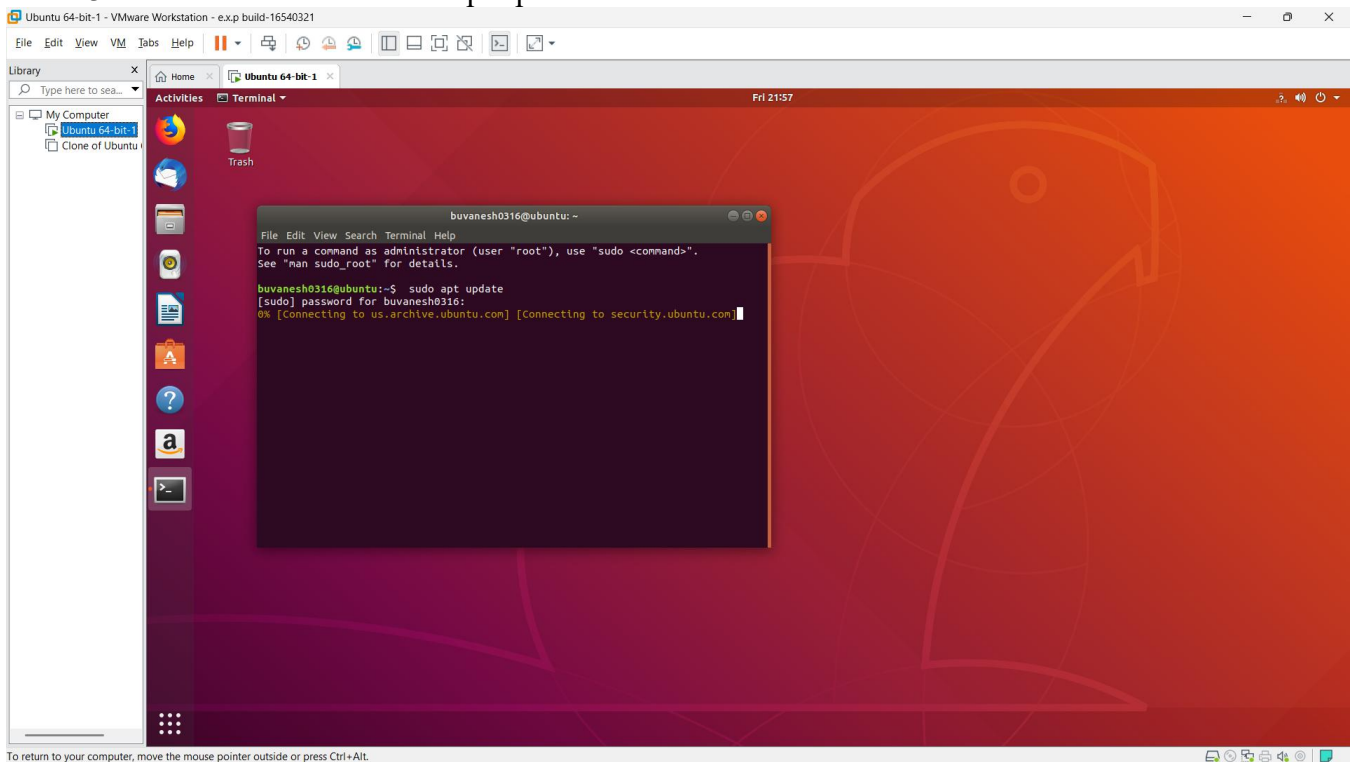
STEP 1: POWER ON THE VIRTUAL MACHINE



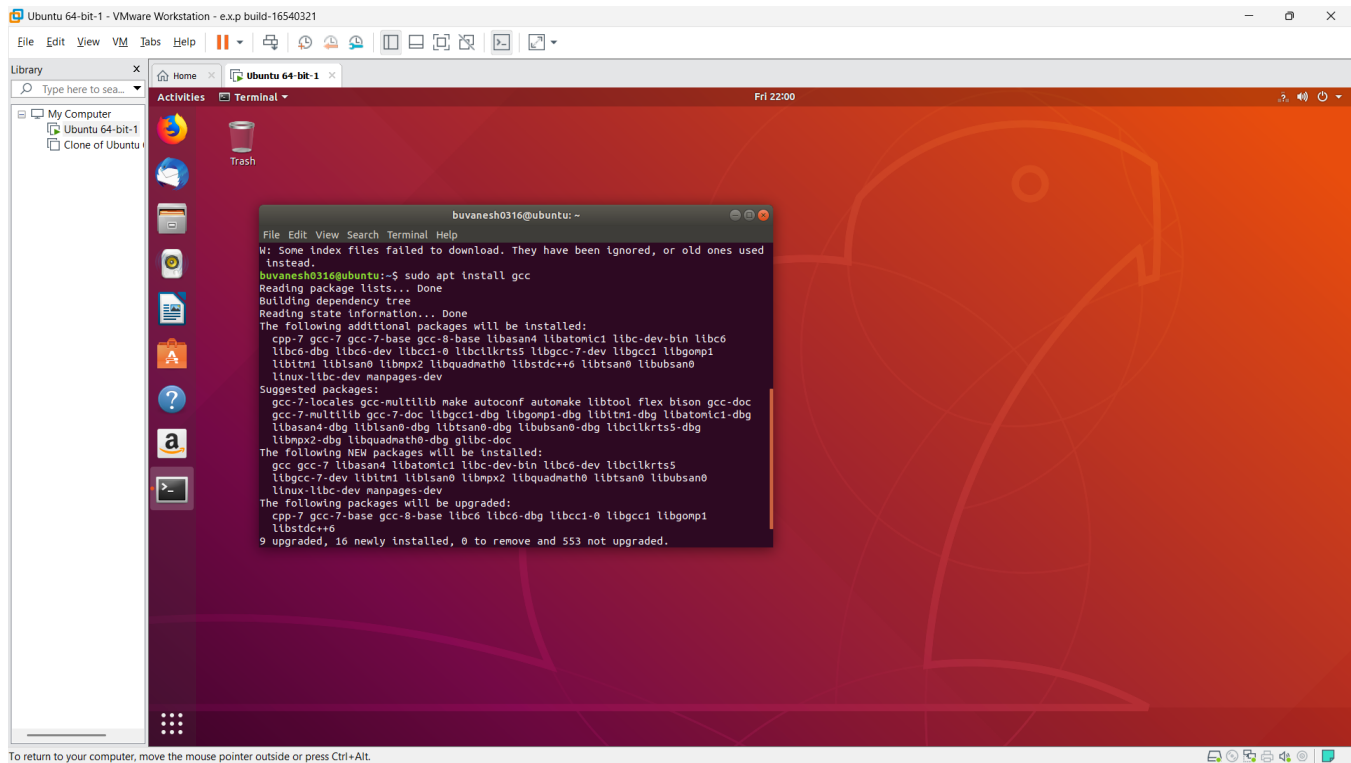
STEP 2: OPEN TERMINAL INSIDE THE VM



STEP 3: RUN COMMAND: sudo apt update



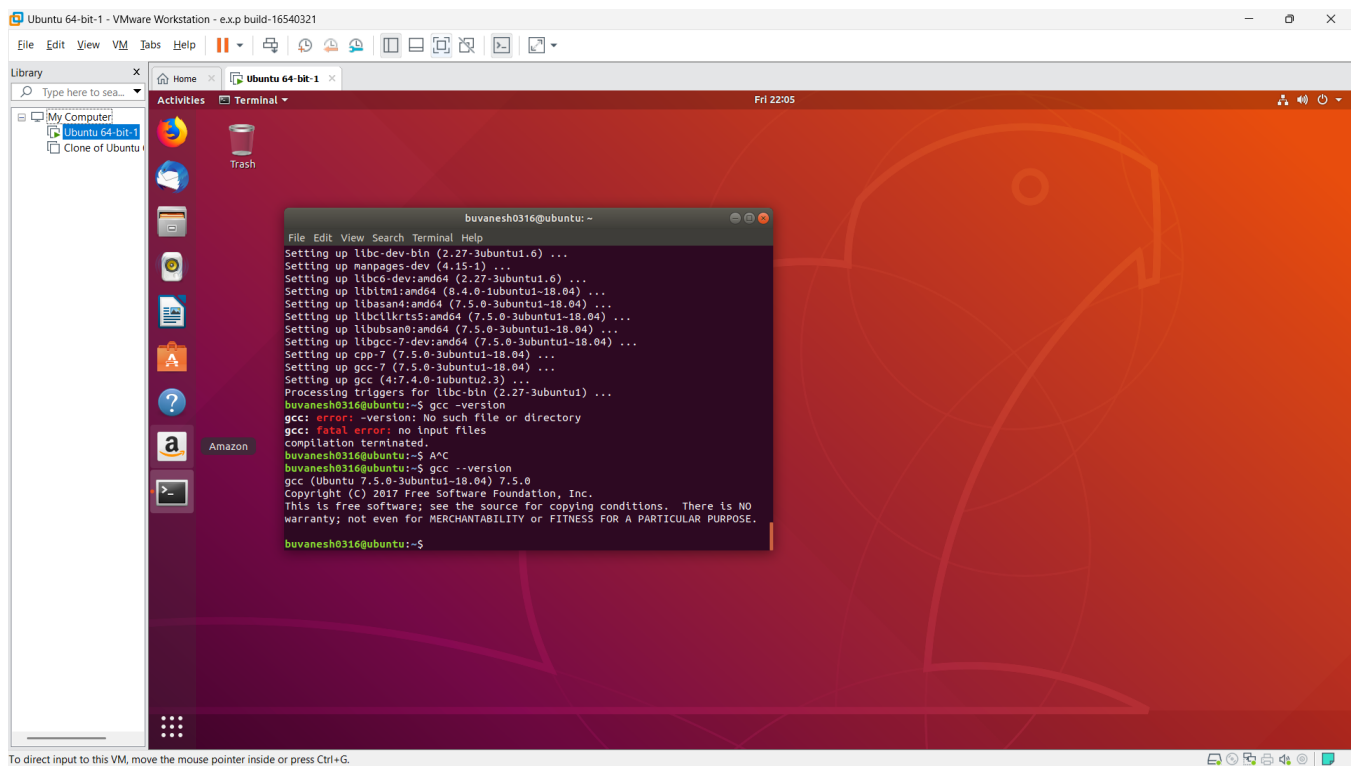
STEP 4: RUN COMMAND: `sudo apt install gcc`



The screenshot shows a terminal window in a VM titled "Ubuntu 64-bit-1". The user has run the command `sudo apt install gcc`. The terminal output shows the following:

```
buvanesh0316@ubuntu: ~  
File Edit View Search Terminal Help  
W: Some index files failed to download. They have been ignored, or old ones used instead.  
buvanesh0316@ubuntu:~$ sudo apt install gcc  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following additional packages will be installed:  
  cpp-7 gcc-7 gcc-7-base gcc-8-base libasan4 libatomic1 libc-dev-bin libc6  
  libc6-dbg libc6-dev libcc1-0 libcilkrts5 libgcc-7-dev libgcc1 libgomp1  
  libitm1 liblsan0 libmpx2 libquadmath0 libstdc++6 libtsan0 libubsan0  
  linux-libc-dev manpages-dev  
Suggested packages:  
  gcc-7-locales gcc-multilib make autoconf automake libtool flex bison gcc-doc  
  gcc-7-multilib gcc-7-doc libgcc1-dbg libgomp1-dbg libitm1-dbg libatomic1-dbg  
  libasan4-dbg liblsan0-dbg libtsan0-dbg libubsan0-dbg libcilkrts5-dbg  
  libmpx2-dbg libquadmath0-dbg glibc-doc  
The following NEW packages will be installed:  
  gcc gcc-7 libasan4 libatomic1 libc-dev-bin libc6-dev libcilkrts5  
  libgcc-7-dev libitm1 liblsan0 libmpx2 libquadmath0 libtsan0 libubsan0  
  linux-libc-dev manpages-dev  
The following packages will be upgraded:  
  cpp-7 gcc-7-base gcc-8-base libc6 libc6-dbg libcc1-0 libgcc1 libgomp1  
  libstdc++6  
9 upgraded, 16 newly installed, 0 to remove and 553 not upgraded.
```

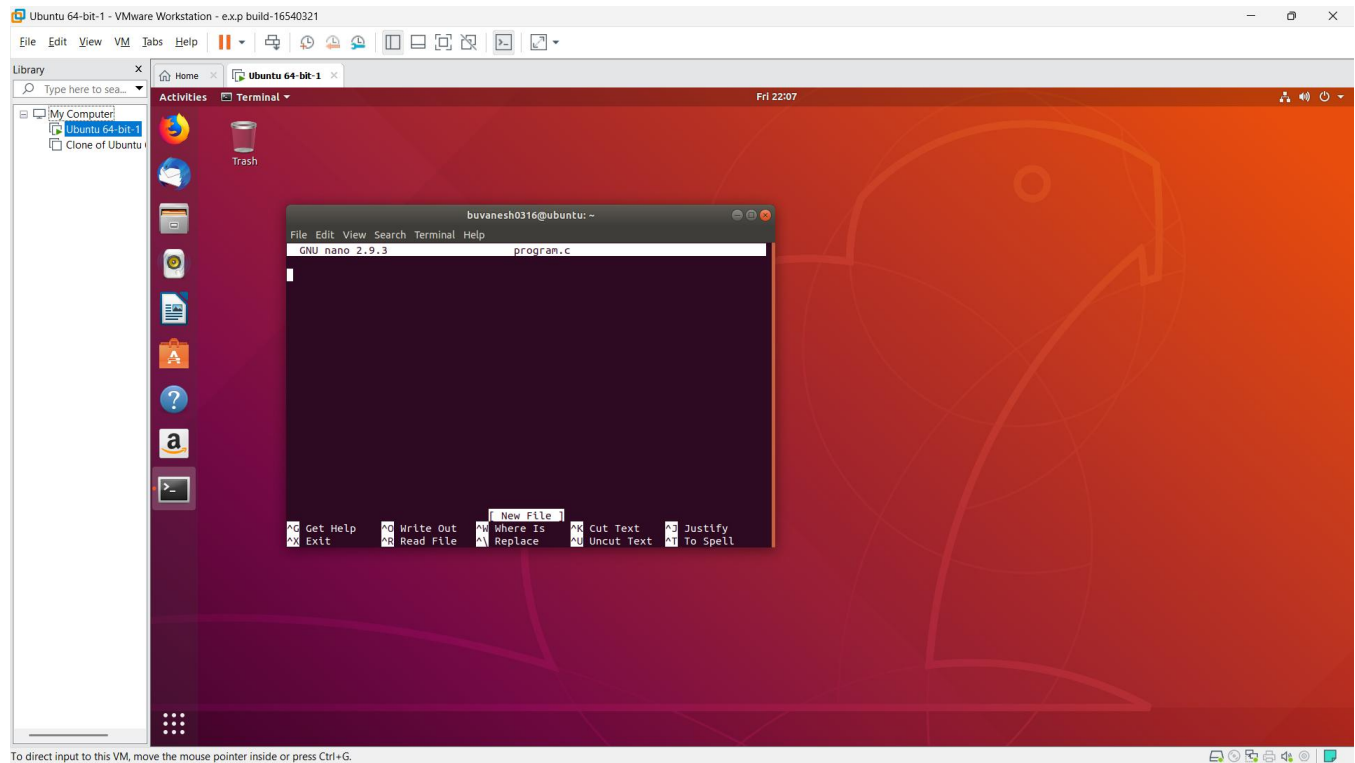
STEP 5: VERIFY INSTALLATION: `gcc --version`



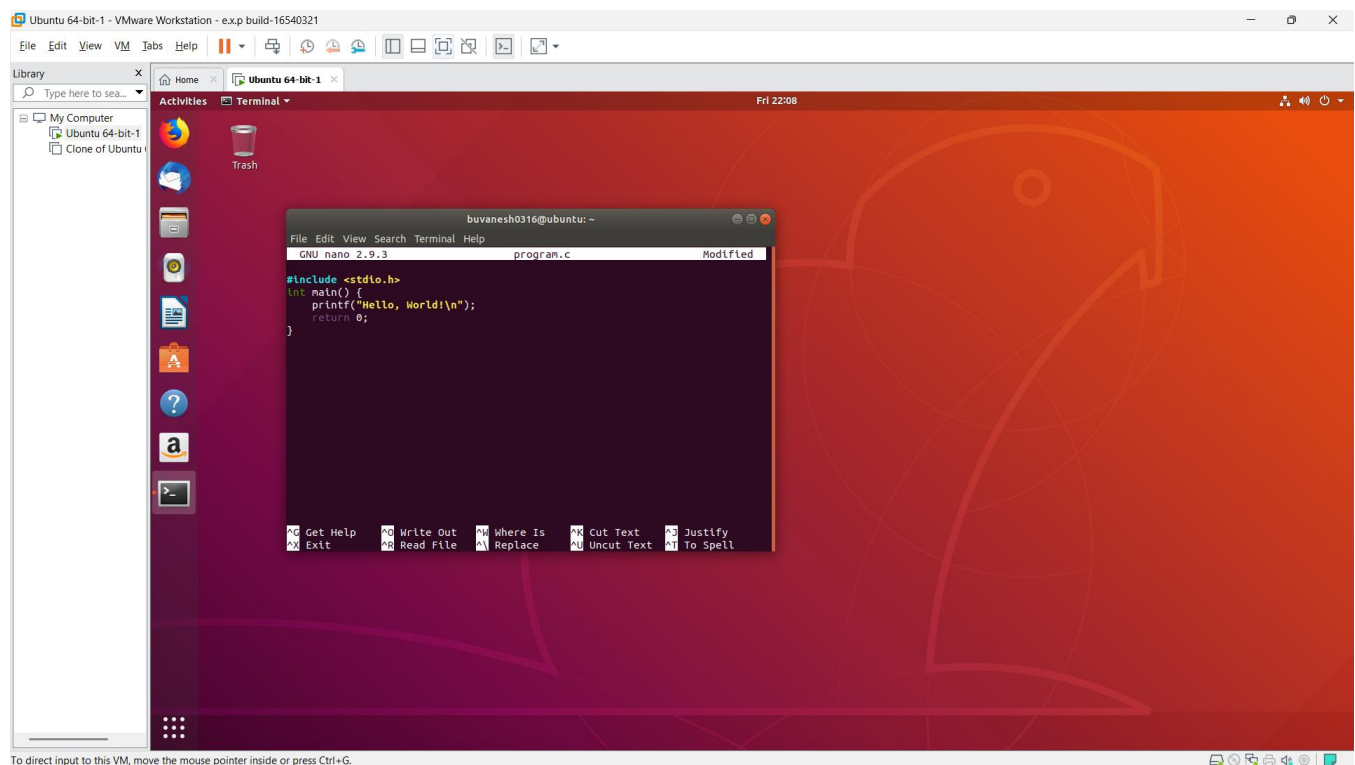
The screenshot shows a terminal window in a VM titled "Ubuntu 64-bit-1". The user has run the command `gcc --version`. The terminal output shows the following:

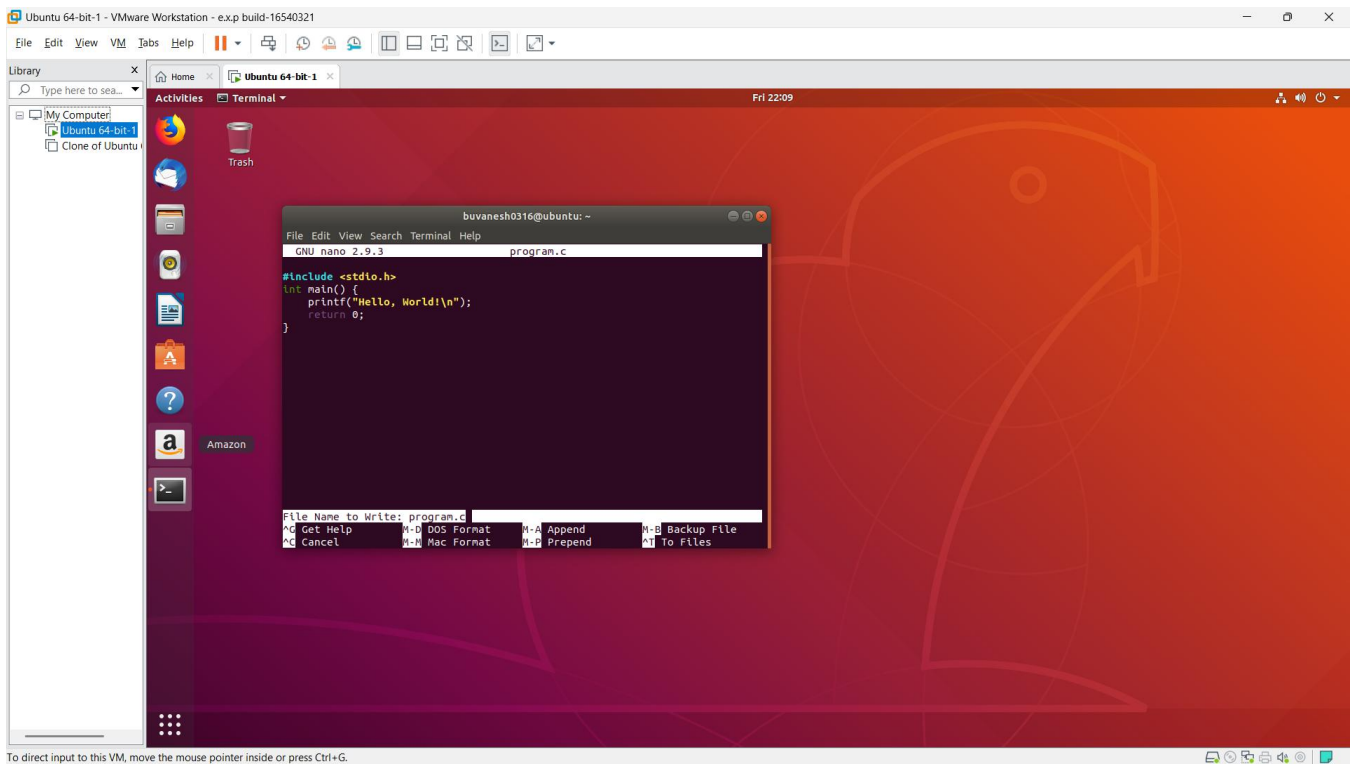
```
buvanesh0316@ubuntu: ~  
File Edit View Search Terminal Help  
Setting up libc-dev-bin (2.27-3ubuntu1.6) ...  
Setting up manpages-dev (4.15-1) ...  
Setting up libc6-dev:amd64 (2.27-3ubuntu1.6) ...  
Setting up libitm1:amd64 (8.4.0-1ubuntu1-18.04) ...  
Setting up libasan4:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libcilkrts5:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libubsan0:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up libgcc-7-dev:amd64 (7.5.0-3ubuntu1-18.04) ...  
Setting up cpp-7 (7.5.0-3ubuntu1-18.04) ...  
Setting up gcc-7 (7.5.0-3ubuntu1-18.04) ...  
Setting up gcc (4:7.4.0-1ubuntu2.3) ...  
Processing triggers for libc-bin (2.27-3ubuntu1) ...  
buvanesh0316@ubuntu:~$ gcc --version  
gcc: error: --version: No such file or directory  
gcc: fatal error: no input files  
compilation terminated.  
buvanesh0316@ubuntu:~$ A^C  
buvanesh0316@ubuntu:~$ gcc --version  
gcc (Ubuntu 7.5.0-3ubuntu1-18.04) 7.5.0  
Copyright (C) 2017 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
buvanesh0316@ubuntu:~$
```

Step 6: Create a C file

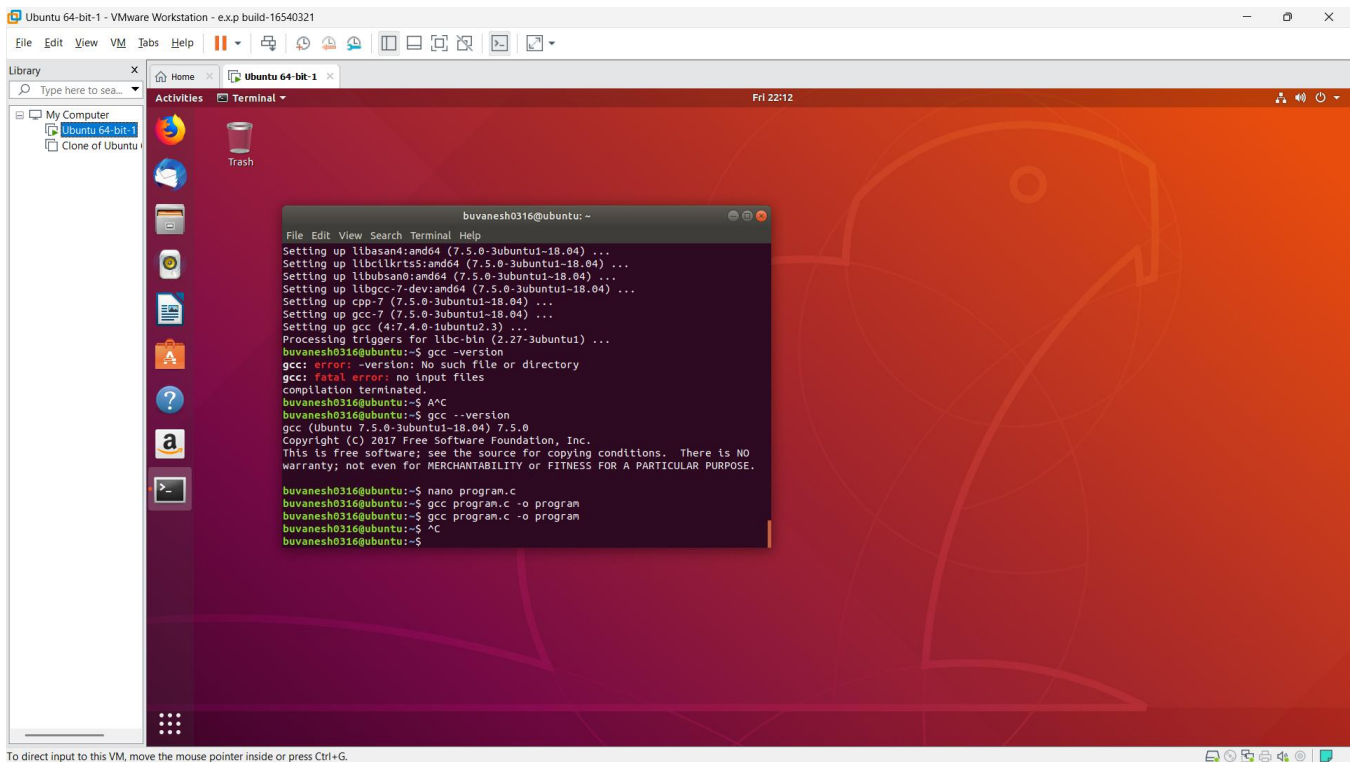


STEP 7: WRITE THE C PROGRAM AND SAVE





STEP 8: COMPILE THE PROGRAM: gcc program.c -o program



STEP 9: RUN THE PROGRAM: ./program

```
buvanesh0316@ubuntu: ~
File Edit View Search Terminal Help
Setting up libasan4:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libcilkrts5:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libubsan0:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libgcc-7-dev:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up cpp-7 (7.5.0-3ubuntu1~18.04) ...
Setting up gcc-7 (7.5.0-3ubuntu1~18.04) ...
Setting up gcc (4:7.4.0-1ubuntu2.3) ...
Processing triggers for libc-bin (2.27-3ubuntu1) ...
buvanesh0316@ubuntu:~$ gcc -version
gcc: error: -version: No such file or directory
gcc: fatal error: no input files
compilation terminated.
buvanesh0316@ubuntu:~$ A^C
buvanesh0316@ubuntu:~$ gcc --version
gcc (Ubuntu 7.5.0-3ubuntu1~18.04) 7.5.0
Copyright (C) 2017 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

buvanesh0316@ubuntu:~$ nano program.c
buvanesh0316@ubuntu:~$ gcc program.c -o program
buvanesh0316@ubuntu:~$ gcc program.c -o program
buvanesh0316@ubuntu:~$ ^C
buvanesh0316@ubuntu:~$ ./program
```

STEP 10: VIEW THE OUTPUT ON TERMINAL

```
buvanesh0316@ubuntu: ~
File Edit View Search Terminal Help
Setting up libubsan0:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up libgcc-7-dev:amd64 (7.5.0-3ubuntu1~18.04) ...
Setting up cpp-7 (7.5.0-3ubuntu1~18.04) ...
Setting up gcc-7 (7.5.0-3ubuntu1~18.04) ...
Setting up gcc (4:7.4.0-1ubuntu2.3) ...
Processing triggers for libc-bin (2.27-3ubuntu1) ...
buvanesh0316@ubuntu:~$ gcc -version
gcc: error: -version: No such file or directory
gcc: fatal error: no input files
compilation terminated.
buvanesh0316@ubuntu:~$ A^C
buvanesh0316@ubuntu:~$ gcc --version
gcc (Ubuntu 7.5.0-3ubuntu1~18.04) 7.5.0
Copyright (C) 2017 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

buvanesh0316@ubuntu:~$ nano program.c
buvanesh0316@ubuntu:~$ gcc program.c -o program
buvanesh0316@ubuntu:~$ gcc program.c -o program
buvanesh0316@ubuntu:~$ ^C
buvanesh0316@ubuntu:~$ ./program
Hello, World!
buvanesh0316@ubuntu:~$
```

Result: Installed and Successfully Ran a C Program VM